Higher-Order Marginal Likelihood Approximation for Nonlinear State Space Models

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Abstract

Bayesian comparison of nonlinear state space models requires integrating over a growing latent state path and a low-dimensional parameter vector. We propose a marginal likelihood approximation that integrates out the states with a numerical or simulation-based evaluator and applies a second-order corrected Laplace approximation to the integral over the static parameters. The correction is expressed through smoothing cumulants of the complete-data log density and, under derivative-growth and posterior-concentration conditions, reduces the log marginal likelihood error from $O(T^{-1})$ to $O(T^{-2})$. In stochastic volatility experiments, it removes 64% to 99% of the Laplace error at far lower cost than simulation-based estimators. Applications to bivariate exchange-rate volatility models and to a sample of $T = 19{,}266$ five-minute stock returns find the strongest support for heavy-tailed observations and for a transitory volatility factor.

JEL classification: C11, C15, C52, C58

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1 Introduction

Nonlinear state space models are widely used to study time-varying volatility, counts, durations, and other latent dynamics in macroeconomics and finance. High-frequency financial data can supply tens of thousands of observations on a single asset, while models of volatility spillovers, time-varying correlations, heavy tails, and jumps can require several interacting latent processes and non-Gaussian observations. Empirical work often requires comparing such competing specifications. These models have two sets of unknowns: a latent state path, whose dimension grows with the sample size T, and a small vector of static parameters $\boldsymbol{\theta}$. Bayesian model comparison is based on the marginal likelihood, which integrates over both. Integrating out the states gives the integrated likelihood $Z_T(\boldsymbol{\theta}) = p(\mathbf{y} \mid \boldsymbol{\theta})$, and integrating this against the prior density $p(\boldsymbol{\theta})$ gives the marginal likelihood, $m_T = \int Z_T(\boldsymbol{\theta}) p(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta}$.

Computing this nested integral accurately is challenging. In nonlinear and non-Gaussian models, $Z_T(\boldsymbol{\theta})$ is generally unavailable analytically and must be evaluated by filtering or simulation over the full sample. Then computing the marginal likelihood adds an outer integration over the static parameters, which is costly to evaluate. Simulation-based methods typically require either repeated evaluations of $Z_T(\boldsymbol{\theta})$ across parameter values or posterior simulation over the parameters and latent states. These approaches can be computationally demanding for long series and models with complex latent dynamics.

We propose an approximation strategy that computes the marginal likelihood in two steps. First, we integrate out the states using a numerical or simulation-based method that exploits the state space structure. Although the number of latent states grows with the sample size T, the number of static parameters remains fixed. Integrating out the states leaves a fixed, low-dimensional integral over the static parameters, which we approximate in the second step using a Laplace expansion. This approximation relies on the posterior mode and local derivatives, reducing the need for costly likelihood evaluations across the parameter space. Under suitable regularity conditions, its accuracy improves as the posterior of the static parameters concentrates with increasing T, making the approach particularly attractive for long samples, where simulation-based integration becomes computationally demanding.

At moderate sample sizes, however, the posterior of the static parameters can exhibit skewness and other departures from normality, limiting the accuracy of the ordinary

Laplace approximation. We therefore include a second-order correction based on the third and fourth derivatives of the log integrated likelihood and the log prior at the posterior mode. This correction accounts for the leading effects of posterior nonnormality while retaining the computational advantages of a local approximation. Once the mode and the required derivatives have been computed, the proposed approach requires neither posterior draws nor importance sampling over the parameters.

Our theoretical analysis characterizes the correction and its error in terms of the integrated likelihood. We express the third and fourth derivatives of its logarithm through expectations and joint cumulants under the smoothing distribution, which links the correction to the latent-state structure. Under explicit derivative-growth and posterior-concentration conditions, our main theorem shows that the correction reduces the approximation error in the log marginal likelihood from order T^{-1} to order T^{-2}. We separate this approximation error from the likelihood-evaluation error and assess numerical errors in the mode and curvature separately. The theory applies to a fixed, low-dimensional parameter vector with a posterior concentrated around a unique interior mode.

The proposed approach integrates out the states with any evaluator that is accurate for the dimension of the state. For a scalar or bivariate state, a numerical-integration filter on a grid (Kitagawa, 1987; Fridman and Harris, 1998) gives the integrated likelihood to machine precision. For a vector state, a particle filter (Andrieu, Doucet, and Holenstein, 2010) gives an unbiased evaluation, and the approach then uses the ordinary Laplace approximation because the filter noise prevents reliable estimation of the higher derivatives. A bounded-weight evaluator, where one exists, gives a finite-sample bound on the simulation error; the rejection sampler of Chan (2026) is one for scalar states, and we use it as an independent check of the grid filter.

In a series of Monte Carlo experiments, we measure the approximation error for stochastic volatility and three related scalar-state models against quadrature benchmarks accurate to 10^{-4}. This setting is chosen so that the marginal likelihood estimators in most common use can be run and compared. In the stochastic volatility experiments, the correction removes 64% to 88% of the Laplace error at $T = 500$ and 94% to 99% at $T = 8{,}000$, leaving log errors below 0.08 and 0.003. At $T = 8{,}000$ and equal computing time, the remaining error is below the numerical standard errors of importance sampling with the cross-entropy method (Chan and Eisenstat, 2015), Chib's method (Chib, 1995; Chib and Jeliazkov, 2001) and the Gelfand–Dey estimator (Gelfand and Dey, 1994; Geweke, 1999)

on the same data sets. Finally, we compare four bivariate stochastic volatility models, with two- or three-dimensional states, on data generated from each of them. The proposed estimator favors the true model in 196 of the 200 series; the four exceptions are near-ties on data from the simplest model, which the other three include as a special case.

We then present two applications. The first compares four bivariate stochastic volatility specifications drawn from Yu and Meyer (2006) using $T = 4{,}000$ daily AUD/USD and NZD/USD returns. Starting from a constant-correlation baseline, the other three models separately introduce volatility spillovers, heavy-tailed observations, and time-varying correlation. Yu and Meyer's DIC-based comparison favors specifications with time-varying correlations on weekly exchange-rate data. We revisit these specifications using marginal likelihoods and a more recent daily sample. To our knowledge, marginal likelihoods have not previously been reported for these four specifications. In our sample, the heavy-tailed model receives the strongest support, with a log Bayes factor of 110 relative to the baseline, followed by time-varying correlation at 104 and volatility spillovers at 98.

The second application compares seven stochastic volatility models for a sample of $T = 19{,}266$ five-minute returns on the S&P 500 ETF over 2025: the standard stochastic volatility model and its extensions with heavy tails, with jumps and with a second volatility factor. The two extensions that change the dynamics of the volatility, a jump in the volatility and a transitory factor, have log Bayes factors of 69 to 73 relative to the baseline, against 12 to 25 for heavy tails and return jumps. The two-factor volatility model is the best supported. It separates a persistent factor with a half-life of about two trading days from a transitory factor with a half-life of nineteen minutes, and once the transitory factor is included, heavy tails and jumps lower the marginal likelihood.

The paper contributes to three strands of literature. The first is simulation-based marginal likelihood estimation: Chib's method (Chib, 1995; Chib and Jeliazkov, 2001), bridge sampling (Meng and Wong, 1996), importance sampling (Geweke, 1989; Chan and Eisenstat, 2015; Chan, 2023) and sequential Monte Carlo (Creal, 2012; Chopin, Jacob, and Papaspiliopoulos, 2013). All of these methods simulate over the parameters; most require posterior draws from MCMC, and the variational importance sampler of Chan, Yu, and Zhang (2026) avoids the chain but still draws the parameters from an importance density. Our contribution is to approximate the integral over the static parameters directly by a Laplace approximation with a second-order correction. This eliminates simulation over the parameters and gives an approximation error of order T^{-2}.

The second is the literature on Laplace approximations and posterior expansions, including applications to Bayesian DSGE models (Schorfheide, 2000). The higher-order correction itself is old: it is the next term of Laplace's method, and Tierney and Kadane (1986), Kass, Tierney, and Kadane (1990) and Shun and McCullagh (1995) develop it for statistical applications. Our contribution is its formulation and error analysis for nonlinear state space models after the latent states have been integrated out. The derivatives that enter the correction can be written as smoothing cumulants, and the approximation error is separated from the numerical errors in evaluating the integrated likelihood and its derivatives. This differs from integrated nested Laplace approximation (Rue, Martino, and Chopin, 2009; Hubin and Storvik, 2016), which approximates the latent-field integral before numerical integration over hyperparameters, and from the higher-order conditional Laplace approach of Ruli, Sartori, and Ventura (2016).

The third is likelihood evaluation for nonlinear state space models. Numerical-integration filters (Kitagawa, 1987; Fridman and Harris, 1998; Farmer, 2021), the exact filter of Creal (2017), and importance samplers (Daníelsson and Richard, 1993; Durbin and Koopman, 1997; Richard and Zhang, 2007) address the inner state integral. We use these methods as building blocks for marginal likelihood approximation and characterize what the integrated-likelihood evaluator must provide for the outer expansion. The smoothing-cumulant representation extends the score and information identities of Louis (1982).

The rest of the paper is organized as follows. Section 2 describes the proposed approach. Section 3 states the theoretical results. Section 4 measures the accuracy of the proposed estimator on generated data and how often it selects the true model. Sections 5 and 6 present the applications, and Section 7 concludes.

2 Marginal Likelihood Approximation

The proposed approach combines an evaluator of the integrated likelihood with a Laplace approximation to the integral over the static parameters. We first formulate this integral in unconstrained coordinates. Then we derive the Laplace approximation and its second-order correction in standardized coordinates, describe two evaluators of the integrated likelihood at the mode together with the bounded-weight check, and decompose the error of the proposed estimator into a simulation term and an approximation term.

Let $\mathbf{y} = (\mathbf{y}_1', \ldots, \mathbf{y}_T')'$ denote the observations and $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_1', \ldots, \boldsymbol{\alpha}_T')'$ a latent state path.

Conditional on a static parameter vector $\boldsymbol{\theta}$, the states form a Markov chain with transition density $p(\boldsymbol{\alpha}_t \,|\, \boldsymbol{\alpha}_{t-1}, \boldsymbol{\theta})$ and initial density $p(\boldsymbol{\alpha}_1 \,|\, \boldsymbol{\theta})$. The observations are conditionally independent given the states, with density $p(\mathbf{y}_t \,|\, \boldsymbol{\alpha}_t, \boldsymbol{\theta})$. The integrated likelihood and the marginal likelihood are, respectively,

$$Z_T(\boldsymbol{\theta}) = \int p(\mathbf{y} \,|\, \boldsymbol{\alpha}, \boldsymbol{\theta})\, p(\boldsymbol{\alpha} \,|\, \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\alpha}, \qquad m_T = \int Z_T(\boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta},$$

where $p(\boldsymbol{\theta})$ is a proper prior. Model comparison needs to use the integrated likelihood Z_T, because criteria computed from the likelihood conditional on the states are unreliable (see, e.g., Chan and Grant, 2015, 2016; Li, Yu, and Zeng, 2020).

Our running example is the stochastic volatility model

$$y_t = \mathrm{e}^{h_t/2}\varepsilon_t, \qquad h_t = \mu + \phi(h_{t-1} - \mu) + \sigma\eta_t, \qquad h_1 \sim \mathcal{N}\big(\mu, \sigma^2/(1-\phi^2)\big), \tag{1}$$

with ε_t and η_t independent standard normal. The state is the scalar log-volatility h_t, and the static parameters are $\boldsymbol{\theta} = (\mu, \phi, \sigma^2)'$, where ϕ and σ have the constrained supports $|\phi| < 1$ and $\sigma > 0$.

A Gaussian approximation, such as that underlying the Laplace method, does not respect these constrained supports. Therefore, we transform the static parameters to unconstrained coordinates to remove parameter boundaries and potentially reduce posterior asymmetry, thereby improving approximation accuracy. Specifically, let $\boldsymbol{\psi} = g(\boldsymbol{\theta}) \in \mathbb{R}^d$ be a smooth bijection, and let $p_{\boldsymbol{\psi}}(\boldsymbol{\psi})$ denote the transformed prior density. In these coordinates, define the log posterior kernel as

$$f(\boldsymbol{\psi}) = \log Z_T(\boldsymbol{\psi}) + \log p_{\boldsymbol{\psi}}(\boldsymbol{\psi}),$$

where $Z_T(\boldsymbol{\psi})$ denotes the integrated likelihood evaluated at $\boldsymbol{\theta} = g^{-1}(\boldsymbol{\psi})$. The marginal likelihood can then be written as

$$m_T = \int_{\mathbb{R}^d} \mathrm{e}^{f(\boldsymbol{\psi})}\, \mathrm{d}\boldsymbol{\psi}.$$

For the stochastic volatility model (1), we use $\boldsymbol{\psi} = (\mu, \operatorname{atanh}\phi, \log\sigma^2)'$, which maps the restricted supports of the persistence parameter $\phi \in (-1, 1)$ and the variance $\sigma^2 > 0$ to the real line. Appendix B records all transformations and their Jacobian terms used in

this paper.

2.1 Laplace Approximation and the Second-Order Correction

Let $\widehat{\boldsymbol{\psi}}$ denote the posterior mode, the maximizer of the log posterior kernel f, and define $\boldsymbol{\Lambda} = -\nabla^2 f(\widehat{\boldsymbol{\psi}})$ as the negative Hessian matrix evaluated at the mode, assumed positive definite, and let $\mathbf{A} = \boldsymbol{\Lambda}^{-1/2}$ be its symmetric inverse square root. The ordinary Laplace approximation replaces the log posterior kernel f by its quadratic expansion around the mode and integrates the resulting Gaussian kernel. This gives the approximation $m_T \approx m_1$, where

$$\begin{aligned} m_1 &= \int_{\mathbb{R}^d} \exp\left\{ f(\widehat{\boldsymbol{\psi}}) - \frac{1}{2}(\boldsymbol{\psi} - \widehat{\boldsymbol{\psi}})' \boldsymbol{\Lambda} (\boldsymbol{\psi} - \widehat{\boldsymbol{\psi}}) \right\} \mathrm{d}\boldsymbol{\psi} \\ &= (2\pi)^{d/2} |\boldsymbol{\Lambda}|^{-1/2} \, \mathrm{e}^{f(\widehat{\boldsymbol{\psi}})} \\ &= (2\pi)^{d/2} |\boldsymbol{\Lambda}|^{-1/2} \, p_{\boldsymbol{\psi}}(\widehat{\boldsymbol{\psi}}) \, Z_T(\widehat{\boldsymbol{\psi}}). \end{aligned} \tag{2}$$

This approximation is exact when f is globally quadratic. Otherwise, its error reflects the terms omitted from this quadratic expansion. To isolate these terms, introduce the standardized coordinates $\mathbf{u} = \mathbf{A}^{-1}(\boldsymbol{\psi} - \widehat{\boldsymbol{\psi}})$, or equivalently $\boldsymbol{\psi} = \widehat{\boldsymbol{\psi}} + \mathbf{A}\mathbf{u}$. Define

$$R(\mathbf{u}) = \underbrace{f(\widehat{\boldsymbol{\psi}} + \mathbf{A}\mathbf{u})}_{\text{log posterior kernel}} - \underbrace{\left[f(\widehat{\boldsymbol{\psi}}) - \tfrac{1}{2}\mathbf{u}'\mathbf{u} \right]}_{\text{quadratic approximation}}$$

as the difference between the log posterior kernel and its quadratic approximation in standardized coordinates. It can be shown that the ratio of the marginal likelihood to its ordinary Laplace approximation is

$$\frac{m_T}{m_1} = \mathbb{E}\big[\mathrm{e}^{R(\mathbf{G})}\big], \tag{3}$$

where $\mathbf{G} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. The correction follows by expanding R and integrating the resulting terms against the standard Gaussian density (Barndorff-Nielsen and Cox, 1989, Chapter 6); Kass, Tierney, and Kadane (1990) use the same expansion, in coordinates rescaled by the square root of the sample size, to establish the validity of posterior expansions. Because the linear term vanishes at the mode and the quadratic term has been removed,

the Taylor expansion starts at order three:

$$R(\mathbf{u}) = C_3(\mathbf{u}) + C_4(\mathbf{u}) + \cdots, \qquad C_k(\mathbf{u}) = \frac{1}{k!} D^k f(\widehat{\boldsymbol{\psi}})[\mathbf{A}\mathbf{u}, \ldots, \mathbf{A}\mathbf{u}],$$

where $D^k f(\widehat{\boldsymbol{\psi}})[\cdot]$ is the kth derivative of f at $\widehat{\boldsymbol{\psi}}$ as a multilinear form. The coefficients of C_3 and C_4 are the standardized derivative tensors

$$a_{ijk} = D^3 f(\widehat{\boldsymbol{\psi}})[\mathbf{A}\mathbf{e}_i, \mathbf{A}\mathbf{e}_j, \mathbf{A}\mathbf{e}_k], \qquad b_{ijkl} = D^4 f(\widehat{\boldsymbol{\psi}})[\mathbf{A}\mathbf{e}_i, \mathbf{A}\mathbf{e}_j, \mathbf{A}\mathbf{e}_k, \mathbf{A}\mathbf{e}_l],$$

where $\mathbf{e}_i$ is the ith column of $\mathbf{I}_d$, so that $\mathbf{A}\mathbf{e}_i$ is the ith column of $\mathbf{A}$. Taking the expectation in (3) term by term, the odd-order terms vanish by the symmetry of the Gaussian, and the leading correction to $m_T/m_1 = 1$ is $\mathbb{E}[C_4(\mathbf{G})] + \frac{1}{2}\mathbb{E}[C_3(\mathbf{G})^2]$. In terms of the tensors, this second-order correction is

$$\Delta_2 = \frac{1}{8} \sum_{i,j} b_{iijj} + \frac{1}{12} \sum_{i,j,k} a_{ijk}^2 + \frac{1}{8} \sum_i \Big(\sum_j a_{ijj} \Big)^2, \tag{4}$$

where the first sum is $\mathbb{E}[C_4(\mathbf{G})]$ and the second and third sums together are $\frac{1}{2}\mathbb{E}[C_3(\mathbf{G})^2]$, the two contractions of the square of the cubic form under the Gaussian sixth moment. In the usual case in which $\boldsymbol{\Lambda}$ grows linearly in T, the standardized tensors satisfy $a_{ijk} = O(T^{-1/2})$ and $b_{ijkl} = O(T^{-1})$, so $\Delta_2 = O(T^{-1})$ and the terms left out are of order T^{-2}. Our main theorem in Section 3 gives conditions under which $\log m_T = \log m_1 + \Delta_2 + O(T^{-2})$. This correction is well known. The three contractions in (4) are the level-one terms of the expansion of Shun and McCullagh (1995) for integrals in d dimensions, enumerated by McCullagh (1987, pp. 254–256). For $d = 1$, (4) reduces to the correction $f^{(4)}(\widehat{\psi})/(8\Lambda^2) + 5f^{(3)}(\widehat{\psi})^2/(24\Lambda^3)$ of Tierney and Kadane (1986), with $\Lambda = -f''(\widehat{\psi})$.

The proposed estimator replaces $Z_T(\widehat{\boldsymbol{\psi}})$ in (2) by an evaluation $\widehat{Z}_T(\widehat{\boldsymbol{\psi}})$ and adds the correction:

$$\log \widehat{m}_2 = \tfrac{d}{2} \log(2\pi) - \tfrac{1}{2} \log |\boldsymbol{\Lambda}| + \log p_{\boldsymbol{\psi}}(\widehat{\boldsymbol{\psi}}) + \log \widehat{Z}_T(\widehat{\boldsymbol{\psi}}) + \Delta_2. \tag{5}$$

We write $\widehat{m}_1$ for the same expression without Δ_2. The subscripts distinguish the ordinary and corrected approximations, while the hats indicate use of an evaluated likelihood.

Implementation combines an integrated-likelihood evaluator with numerical optimization and derivative calculations. These calculations use the log integrated likelihood and the

log prior, following the numerical procedures described in Section 2.3. The next subsection discusses the choice of integrated-likelihood evaluator.

2.2 Evaluating the Integrated Likelihood at the Mode

Several established filtering and simulation methods can compute the integrated likelihood. Numerical-integration filters propagate the filtering density on a grid (Kitagawa, 1987; Fridman and Harris, 1998; Watanabe, 1999; Langrock, MacDonald, and Zucchini, 2012; Bégin and Boudreault, 2021; Farmer, 2021). Exact filters exist for special structures, such as the noncentral-gamma state dynamics of Creal (2017). Importance samplers for the state path (Daníelsson and Richard, 1993; Shephard and Pitt, 1997; Durbin and Koopman, 1997; Richard and Zhang, 2007; McCausland, 2012; Chan, 2023) and particle filters (Andrieu, Doucet, and Holenstein, 2010; Pitt, dos Santos Silva, Giordani, and Kohn, 2012, and Creal 2012 for a survey) give simulation estimates. These are unbiased for the particle filter and for importance samplers whose importance density does not depend on the draws used in the estimate; the efficient importance sampler of Richard and Zhang (2007), for instance, fits its density to the same common random numbers.

The proposed approach requires an evaluator that is accurate at one point and, for the second-order correction, has noise small enough for the derivatives to be computed. We use two evaluators, chosen by the dimension of the state, and a third construction that gives a finite-sample bound on the simulation error where it applies.

The first evaluator is for a scalar or bivariate state, where the forward filter can be run on a grid. For a scalar state, let $\alpha_i = \alpha_{\min} + i\,\delta$, $i = 1, \ldots, N$, be equally spaced points. Let $\mathbf{K}$ be the $N \times N$ matrix with entries $K_{ij} = p(\alpha_i \,|\, \alpha_j, \boldsymbol{\theta})\,\delta$, the transition density between grid points times the trapezoidal weight δ, and let $\mathbf{D}_t = \operatorname{diag}\big(p(\mathbf{y}_t \,|\, \alpha_1, \boldsymbol{\theta}), \ldots, p(\mathbf{y}_t \,|\, \alpha_N, \boldsymbol{\theta})\big)$ be the diagonal matrix of the observation densities at the grid points. Further define

$$\mathbf{a}_1 = \mathbf{D}_1 \left(p(\alpha_1 \,|\, \boldsymbol{\theta}), \ldots, p(\alpha_N \,|\, \boldsymbol{\theta})\right)', \qquad \mathbf{a}_t = \mathbf{D}_t \mathbf{K} \mathbf{a}_{t-1}, \quad t = 2, \ldots, T,$$

so that $\mathbf{a}_t$ approximates the joint density $p(\mathbf{y}_1, \ldots, \mathbf{y}_t, \alpha_t = \alpha_i)$ of the observations to date and the state at the grid points, and $Z_T(\boldsymbol{\theta}) \approx \delta\, \mathbf{1}'\mathbf{a}_T$. To avoid underflow, each $\mathbf{a}_t$ is divided by $c_t = \delta\, \mathbf{1}'\mathbf{a}_t$, the one-step-ahead predictive density $p(\mathbf{y}_t \,|\, \mathbf{y}_1, \ldots, \mathbf{y}_{t-1}, \boldsymbol{\theta})$, after which $\mathbf{a}_t$ is the filtering density at the grid points and $\log Z_T(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log c_t$. This is the numerical-integration filter of Kitagawa (1987), the grid-based filter of the nonlinear

filtering literature; we call it the grid filter.

The filter is accurate because the integrand at each step is analytic and decays in both tails. For such an integrand, the error of the trapezoidal rule decays exponentially as the spacing δ shrinks (Trefethen and Weideman, 2014), so a spacing of a fraction of the width of the transition and observation densities gives $\log Z_T$ to machine precision. We provide further details of the grid design and accuracy checks in Appendix B. Each step requires a multiplication by $\mathbf{K}$, so the cost is linear in T and grows as the square of the number of grid points per dimension; the filter is therefore limited to scalar and bivariate states. To our knowledge, it has not been used for marginal likelihood evaluation.

The second evaluator is for a vector state, where a particle filter with P particles gives an unbiased estimate of $Z_T(\widehat{\boldsymbol{\psi}})$. Its variance at the mode is estimated from independent runs, and the estimate enters (5) in the same way.

The particle filter is unbiased, but its simulation error is known only through its estimated variance. The third construction is a simulation evaluator whose error has a finite-sample bound: an importance sampler for the state path whose weights are bounded by a computable constant. For a scalar state with an affine Gaussian transition and an observation log density that is concave in the state, the construction of Chan (2026) gives such a proposal $q_{\widehat{\boldsymbol{\psi}}}(\boldsymbol{\alpha})$ for the state path, built by twisting the transition densities with tangent-line envelopes of concave backward functions, each the observation log density plus a log normalizing constant, together with a computable constant $C(\widehat{\boldsymbol{\psi}})$. The importance weight $w(\boldsymbol{\alpha}) = p(\mathbf{y} \mid \boldsymbol{\alpha}, \widehat{\boldsymbol{\psi}})p(\boldsymbol{\alpha} \mid \widehat{\boldsymbol{\psi}})/q_{\widehat{\boldsymbol{\psi}}}(\boldsymbol{\alpha})$ satisfies $0 < w(\boldsymbol{\alpha}) \leqslant C(\widehat{\boldsymbol{\psi}})$ for every path. With M independent proposals $\boldsymbol{\alpha}^{(1)}, \ldots, \boldsymbol{\alpha}^{(M)}$ and $U_m = w(\boldsymbol{\alpha}^{(m)})/C(\widehat{\boldsymbol{\psi}}) \in (0, 1]$,

$$\widehat{Z}_T(\widehat{\boldsymbol{\psi}}) = C(\widehat{\boldsymbol{\psi}})\,\bar{U}_M, \qquad \bar{U}_M = \frac{1}{M}\sum_{m=1}^{M} U_m, \tag{6}$$

is unbiased for $Z_T(\widehat{\boldsymbol{\psi}})$, and $\mathbb{E}[U_m] = Z_T(\widehat{\boldsymbol{\psi}})/C(\widehat{\boldsymbol{\psi}})$ is the acceptance probability of the rejection sampler built from the same proposal and bound. We call (6) the bounded-weight evaluator. It is built from the rejection sampler of Chan (2026): the normalized weights U_m are the acceptance probabilities of that sampler, and the evaluator averages them in place of drawing accept-or-reject decisions, which removes the rejection noise. Because the weights are bounded, the relative variance of $\widehat{Z}_T(\widehat{\boldsymbol{\psi}})$ is at most $(1/p_{\mathrm{a}} - 1)/M$ at acceptance probability p_{a}, with no condition on the realized weights to verify, in contrast with the importance samplers with Gaussian proposals, whose weight variance

has to be tested (Koopman, Shephard, and Creal, 2009). The class covers stochastic volatility with Gaussian or Student-t errors, stochastic volatility in mean, Poisson and probit observations, and stochastic conditional duration. We use this evaluator in the Monte Carlo experiments as an independent check of the grid filter and to show how small the simulation error of a bounded-weight evaluation is in finite samples.

2.3 Mode Finding and Error Decomposition

This subsection describes how we compute the posterior mode $\widehat{\boldsymbol{\psi}}$, the curvature $\boldsymbol{\Lambda}$ and, for $\widehat{m}_2$, the third and fourth derivatives in (4), and how the errors of these quantities and of the likelihood evaluation enter the proposed estimator. With the grid filter, $\log Z_T$ is deterministic and can be evaluated to an accuracy of 10^{-11}. We use Newton–Raphson iterations in the standardized coordinates $\mathbf{u}$, with gradients and Hessians computed by central differences. The negative Hessian at the mode is $\boldsymbol{\Lambda}$. The correction (4) needs only the third derivatives a_{ijk} and the fourth derivatives of the form b_{iijj}, which we compute by central differences on a sparse design in $\mathbf{u}$. The whole computation requires a few hundred to a few thousand evaluations of $\log Z_T$. The evaluations within each Newton iteration and within the derivative calculation are independent, so they run in parallel batches. With enough cores, the wall-clock cost is a few dozen evaluation times. Appendix B provides the implementation details and the evaluation counts.

When Z_T is evaluated by the bounded-weight evaluator or a particle filter, $\log \widehat{Z}_T$ is noisy and direct numerical differentiation is unreliable. We instead evaluate $\log \widehat{Z}_T$ plus the log prior on a design around a preliminary mode, with common random numbers across the design points, and fit a local polynomial surface.[1] The fitted surface gives $\widehat{\boldsymbol{\psi}}$ and $\boldsymbol{\Lambda}$. With the bounded-weight evaluator, precise enough evaluations also support the estimation of the higher-order derivatives. The noise of a particle filter is in general too large for this purpose, so in that case we use the ordinary Laplace estimator $\widehat{m}_1$.

[1] The preliminary mode is a plug-in start from fits of simpler models. For example, for the bivariate stochastic volatility models in Section 5, each volatility process starts at the posterior mode of a univariate stochastic volatility model fitted to its own series, and the correlation starts at the sample correlation. The search for the dynamic-correlation model starts at the mode of the constant-correlation model, with the correlation process at its correlation, a persistence of 0.95 and an innovation variance of 0.01.

Given $\widehat{\boldsymbol{\psi}}$, $\boldsymbol{\Lambda}$, and Δ_2, the error of the proposed estimator decomposes exactly as

$$\log \widehat{m}_2 - \log m_T = \underbrace{\log \widehat{Z}_T(\widehat{\boldsymbol{\psi}}) - \log Z_T(\widehat{\boldsymbol{\psi}})}_{\text{simulation}} + \underbrace{\Delta_2 - \log \mathbb{E}\left[\mathrm{e}^{R(\mathbf{G})}\right]}_{\text{approximation}}. \tag{7}$$

With the grid filter, the simulation term is zero to machine precision, and with a particle filter its standard error is estimated from independent runs. With the bounded-weight evaluator, its standard error is

$$\mathrm{NSE}(\log \widehat{m}_2) = \frac{s_U}{\bar{U}_M \sqrt{M}},$$

where s_U is the sample standard deviation of $U_1, \ldots, U_M$. Since the U_m are independent and bounded, the normal approximation is reliable for the acceptance probabilities and sample sizes used here; the next section also gives a finite-sample bound for this term.

The approximation term is deterministic given the data, and the next section shows that under regularity conditions on f it is of order T^{-2}. The decomposition in (7) is valid for any $\widehat{\boldsymbol{\psi}}$ and positive definite $\boldsymbol{\Lambda}$, with m_1 and R defined by (2) and (3) at those values. With the grid filter, an inexact mode or curvature therefore enters through the approximation term and adds no separate source of randomness. With a noisy evaluator, the fitted mode and curvature depend on the simulation draws, so the approximation term varies with them as well, and the standard error above covers the final evaluation only. The rate T^{-2} describes the analytic expansion at the exact mode, curvature and derivatives; the implemented estimator attains it only when these are computed to that accuracy, which the extrapolated finite differences do.

The same expansion also clarifies the behavior of importance sampling based on the Laplace proposal. With M draws $\mathbf{G}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, the average of $\mathrm{e}^{R(\mathbf{G}^{(m)})}$ is an unbiased estimator of m_T/m_1 in (3). Its variance is $\mathrm{Var}(\mathrm{e}^{R(\mathbf{G})})/M$. Under additional tail conditions controlling the second moment,

$$\mathrm{Var}(\mathrm{e}^{R(\mathbf{G})}) = \mathbb{E}[C_3(\mathbf{G})^2] + O(T^{-2}),$$

where the leading term is twice the cubic contribution to Δ_2. These tail conditions are stronger than the conditions for the expansion itself. When the variance expansion holds, the standard error of the log importance-sampling estimate is of order $(TM)^{-1/2}$, compared with the T^{-2} approximation error of $\widehat{m}_2$ at a fixed number of evaluations. In

the Monte Carlo experiments we compare the two at equal computing time, with the Gaussian proposal of the cross-entropy method of Chan and Eisenstat (2015).

3 Theoretical Results

We provide the theoretical results behind the two terms of the error decomposition in (7). The first subsection treats the approximation term: Proposition 1 expresses the derivatives of the log integrated likelihood that enter the second-order term as cumulants under the smoothing distribution of the states, and Theorem 1 states the conditions under which the corrected approximation has an error of order T^{-2}. The second subsection treats the simulation term: Proposition 2 gives a finite-sample bound on the error of a bounded-weight evaluation at the mode.

3.1 The Approximation Term

Let $\mathcal{C}(\boldsymbol{\psi}, \boldsymbol{\alpha}) = \log p(\mathbf{y} \,|\, \boldsymbol{\alpha}, \boldsymbol{\psi}) + \log p(\boldsymbol{\alpha} \,|\, \boldsymbol{\psi})$ denote the complete-data log density, so that $Z_T(\boldsymbol{\psi}) = \int \mathrm{e}^{\mathcal{C}(\boldsymbol{\psi}, \boldsymbol{\alpha})} \, \mathrm{d}\boldsymbol{\alpha}$. Let $p(\boldsymbol{\alpha} \,|\, \mathbf{y}, \boldsymbol{\psi})$ denote the smoothing distribution. For a fixed direction $\mathbf{w} \in \mathbb{R}^d$, write $c_j = D^j_{\mathbf{w}} \, \mathcal{C}(\boldsymbol{\psi}, \boldsymbol{\alpha})$ for the jth directional derivative in $\boldsymbol{\psi}$ at fixed $\boldsymbol{\alpha}$. Write $\mathbb{E}_{\boldsymbol{\psi}}$, $\mathrm{Var}_{\boldsymbol{\psi}}$, $\mathrm{Cov}_{\boldsymbol{\psi}}$ and $\mathrm{cum}_{\boldsymbol{\psi}}$ for moments and joint cumulants under $p(\boldsymbol{\alpha} \,|\, \mathbf{y}, \boldsymbol{\psi})$. The third-order joint cumulants are central moments, $\mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_2) = \mathbb{E}_{\boldsymbol{\psi}}[(c_1 - \mathbb{E}_{\boldsymbol{\psi}} c_1)^2 (c_2 - \mathbb{E}_{\boldsymbol{\psi}} c_2)]$, and $\mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_1, c_1) = \mathbb{E}_{\boldsymbol{\psi}}[(c_1 - \mathbb{E}_{\boldsymbol{\psi}} c_1)^4] - 3\mathrm{Var}_{\boldsymbol{\psi}}(c_1)^2$. Differentiating Z_T under the integral sign gives the score identity $D_{\mathbf{w}} \log Z_T = \mathbb{E}_{\boldsymbol{\psi}}[c_1]$ and the observed-information identity of Louis (1982), $D^2_{\mathbf{w}} \log Z_T = \mathbb{E}_{\boldsymbol{\psi}}[c_2] + \mathrm{Var}_{\boldsymbol{\psi}}(c_1)$. The next result extends them to the third and fourth derivatives.

Proposition 1 (Smoothing cumulants)**.** *For a fixed $\boldsymbol{\psi}$, suppose that $\mathcal{C}$ is four times continuously differentiable in its first argument, and that $\mathbb{E}_{\boldsymbol{\psi}'} \exp(\sum_{j=1}^4 t_j c_j)$, with the c_j evaluated at $\boldsymbol{\psi}'$, is finite for $(\mathbf{t}, \boldsymbol{\psi}')$ in a neighborhood of $(\mathbf{0}, \boldsymbol{\psi})$ and can be differentiated under the integral sign there. Then, at $\boldsymbol{\psi}$,*

$$
\begin{aligned}
D^3_{\mathbf{w}} \log Z_T &= \mathbb{E}_{\boldsymbol{\psi}}[c_3] + 3\,Cov_{\boldsymbol{\psi}}(c_1, c_2) + cum_{\boldsymbol{\psi}}(c_1, c_1, c_1), \\
D^4_{\mathbf{w}} \log Z_T &= \mathbb{E}_{\boldsymbol{\psi}}[c_4] + 4\,Cov_{\boldsymbol{\psi}}(c_1, c_3) + 3\,Var_{\boldsymbol{\psi}}(c_2) + 6\,cum_{\boldsymbol{\psi}}(c_1, c_1, c_2) \\
&\qquad + cum_{\boldsymbol{\psi}}(c_1, c_1, c_1, c_1).
\end{aligned} \tag{8}
$$

The identities in (8) follow from the general rule for the derivative of a joint cumulant (Skovgaard, 1986),

$$D_{\mathbf{w}}\mathrm{cum}_{\boldsymbol{\psi}}(g_1,\ldots,g_k)=\sum_{i=1}^{k}\mathrm{cum}_{\boldsymbol{\psi}}(g_1,\ldots,D_{\mathbf{w}}g_i,\ldots,g_k)+\mathrm{cum}_{\boldsymbol{\psi}}(g_1,\ldots,g_k,c_1), \tag{9}$$

which for $k=1$ is $D_{\mathbf{w}}\mathbb{E}_{\boldsymbol{\psi}}[g]=\mathbb{E}_{\boldsymbol{\psi}}[D_{\mathbf{w}}g]+\mathrm{Cov}_{\boldsymbol{\psi}}(g,c_1)$. We prove the proposition in Appendix A.

We assume a standard regularity condition of Laplace's method, the counterpart of a unique, interior maximum likelihood estimator with a nonsingular information matrix. In the unconstrained coordinates $\boldsymbol{\psi}$ a maximum is interior whenever it exists.

Assumption 1 (Mode)**.** *The function f is twice continuously differentiable, attains its maximum on $\mathbb{R}^d$ only at $\widehat{\boldsymbol{\psi}}$, and $\boldsymbol{\Lambda}=-\nabla^2 f(\widehat{\boldsymbol{\psi}})$ is positive definite.*

The next assumption is stated on the set $\mathcal{N}_T=\{\widehat{\boldsymbol{\psi}}+\mathbf{A}\mathbf{u}:\|\mathbf{u}\|\leqslant\varrho\sqrt{T}\}$ for a fixed $\varrho>0$.

Assumption 2 (Derivatives)**.** *The function f is six times continuously differentiable on an open set containing $\mathcal{N}_T$, and there are constants $K_3,\ldots,K_6$ not depending on T, with $\varrho K_3\leqslant 3/2$, such that, for $k=3,\ldots,6$,*

$$\sup_{\boldsymbol{\psi}\in\mathcal{N}_T}\sup_{\|\mathbf{w}\|=1}\left|D^k f(\boldsymbol{\psi})[\mathbf{A}\mathbf{w},\ldots,\mathbf{A}\mathbf{w}]\right|\leqslant K_k\,T^{1-k/2}.$$

Assumption 2 fixes the rates at which the derivatives of f grow with T, in the standardized coordinates in which the Laplace approximation is a standard Gaussian. Along any unit direction the kth standardized derivative is of order $T^{1-k/2}$: the third is $O(T^{-1/2})$, the fourth $O(T^{-1})$, and the fifth and sixth $O(T^{-3/2})$ and $O(T^{-2})$. These are the rates of a log-likelihood built from T weakly dependent observations, and they make the expansion of R in (3) an expansion in powers of $T^{-1/2}$, with $C_k=O(T^{1-k/2})$, $\Delta_2=O(T^{-1})$ and a remainder of order T^{-2}. The bounds are required uniformly on $\mathcal{N}_T$ because the proof uses them both on the ball in $\mathbf{u}$ where it expands f and on the shell between that ball and $\mathcal{N}_T$.

The rates hold when the information grows linearly in T. If the eigenvalues of $\boldsymbol{\Lambda}$ are of order T, then $\|\mathbf{A}\|$ is of order $T^{-1/2}$, $\mathcal{N}_T$ has a fixed size in the original coordinates, and the assumption follows when the unstandardized derivatives of f up to order six are of

order T there. For the likelihood part of f these derivatives are the smoothing cumulants of Proposition 1 and their analogues up to order six, which are of order T when the derivatives c_j have moments bounded uniformly on $\mathcal{N}_T$ and the smoothing distribution forgets the past at a geometric rate, that is, when the dependence between the states at dates s and t decays like $\rho^{|t-s|}$ for some $\rho < 1$; the prior does not depend on T and its log density is smooth, so it contributes derivatives that are bounded on $\mathcal{N}_T$. The condition $\varrho K_3 \leqslant 3/2$ is a choice of ϱ, since K_3 cannot increase when ϱ decreases; the price is that the next assumption is then required on a smaller set.

Assumption 3 (Concentration)**.** *There is a constant C_0 not depending on T such that $\int_{\mathcal{N}_T^c} e^{f(\boldsymbol{\psi})}\, d\boldsymbol{\psi} \leqslant C_0\, m_1\, T^{-2}$ for all large T.*

Assumption 3 is a weak form of posterior concentration. Under the Laplace regularity conditions of Kass, Tierney, and Kadane (1990) it holds with an exponential rate in place of T^{-2}.

Theorem 1 (Second-order expansion)**.** *Under Assumptions 1–3,*

$$\log m_T = \log m_1 + \Delta_2 + O(T^{-2}),$$

with Δ_2 as in (4)*, and $\Delta_2 = O(T^{-1})$. The implied constant and the threshold on T depend only on d, ϱ, $K_3, \dots, K_6$ and C_0.*

We prove the theorem in Appendix A. The proof expands R to sixth order on a ball of radius $T^{1/8}$ in $\mathbf{u}$, where the Gaussian expectation of the expansion gives Δ_2 and all other terms are of order T^{-2}. It then bounds the contributions of the shell between that ball and $\mathcal{N}_T$, and of $\mathcal{N}_T^c$. The argument follows Kass, Tierney, and Kadane (1990), with the regularity conditions written in standardized coordinates.

3.2 The Simulation Term

The simulation term in (7) is zero for the grid filter. For the bounded-weight evaluator, the following result gives its variance and a finite-sample bound on it, both in terms of the acceptance probability p_{a} and the number of proposals M. The bound is stated through the relative entropy between two Bernoulli distributions,

$$\mathrm{kl}(p\|q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q},$$

with the conventions $0 \log 0 = 0$ and $\mathrm{kl}(p\|q) = \infty$ for $q \in \{0, 1\}$, $p \neq q$.

Proposition 2 (Bounded-weight evaluation)**.** *Let $U_1, \ldots, U_M$ be the normalized weights in* (6)*, independent with values in $(0, 1]$ and mean $p_{\mathrm{a}} = Z_T(\widehat{\boldsymbol{\psi}})/C(\widehat{\boldsymbol{\psi}})$. Then $\widehat{Z}_T(\widehat{\boldsymbol{\psi}})$ is unbiased with $Var(\widehat{Z}_T(\widehat{\boldsymbol{\psi}}))/Z_T(\widehat{\boldsymbol{\psi}})^2 \leqslant (1/p_{\mathrm{a}} - 1)/M$, and for any $\gamma \in (0, 1)$,*

$$\mathbb{P}\Big\{\mathrm{kl}(\bar{U}_M\|p_{\mathrm{a}}) \leqslant \frac{\log(2/\gamma)}{M}\Big\} \geqslant 1 - \gamma.$$

The first part is Corollary 1 of Chan (2026). The second is the relative-entropy form of the inequality of Hoeffding (1963) for bounded variables. Because $a \mapsto \mathrm{kl}(\bar{U}_M\|a)$ is convex with minimum zero at $a = \bar{U}_M$, the set $\{a \in [0, 1] : \mathrm{kl}(\bar{U}_M\|a) \leqslant \log(2/\gamma)/M\}$ is an interval, and by the proposition it contains p_{a} with probability at least $1-\gamma$ for every M; it equals $[\mathrm{e}^{-\log(2/\gamma)/M}, 1]$ when $\bar{U}_M = 1$. Its logarithm shifted by $\log C(\widehat{\boldsymbol{\psi}})$ contains $\log Z_T(\widehat{\boldsymbol{\psi}})$ with the same probability, which bounds the simulation error of $\log \widehat{Z}_T(\widehat{\boldsymbol{\psi}})$. When Mp_{a} is large relative to $\log(2/\gamma)$, the interval has width about $2\{2\bar{U}_M(1 - \bar{U}_M)\log(2/\gamma)/M\}^{1/2}$. Its relative width is therefore of order $(Mp_{\mathrm{a}})^{-1/2}$, and a fixed relative accuracy needs M of order $1/p_{\mathrm{a}}$.

4 Accuracy and Model Selection on Generated Data

In this section we measure the accuracy of the proposed approach and compare it with simulation-based estimators in common use. Accuracy can be measured only against a benchmark, so we use models whose log marginal likelihood can be computed by quadrature. We describe the design and the benchmark, report the error of the proposed estimator across sample sizes, persistence levels and models, and compare it with importance sampling and two MCMC-based estimators at equal computing time. Finally, we generate data from four bivariate stochastic volatility models, which we also compare in the exchange-rate application, and record how often the true model has the largest estimated marginal likelihood.

4.1 Design and Benchmark

We consider four commonly used nonlinear or non-Gaussian state space models. The first is the stochastic volatility model in (1). Three related models change only the observation

equation. The Student-t model has $\varepsilon_t \sim \mathcal{T}_\nu$ with unit scale. The stochastic volatility in mean model (Koopman and Hol Uspensky, 2002; Chan, 2017) has $y_t = \lambda \mathrm{e}^{h_t} + \mathrm{e}^{h_t/2}\varepsilon_t$. The Poisson model has counts $y_t \sim \text{Poisson}(\mathrm{e}^{h_t})$.

Data are generated from the stochastic volatility model with $\mu = 0$ and (ϕ, σ) equal to $(0.90, 0.35)$, $(0.95, 0.25)$ or $(0.98, 0.15)$. The related models use $(\phi, \sigma) = (0.95, 0.25)$, with $\nu = 5$, $\lambda = 0.5$ and, for the Poisson model, $\mu = 2$ and $\sigma = 0.22$. We consider five sample sizes, $T \in \{500, 1{,}000, 2{,}000, 4{,}000, 8{,}000\}$, and take the first T observations of each generated path, so the samples are nested. We generate ten replications for each of the six model and parameter combinations, and the five sample sizes then give 300 data sets. The replications show the size of the approximation error and its dependence on T. The ranges reported below are ranges over replications, not confidence intervals.

The benchmark value of $\log m_T$ for each data set comes from a nested quadrature over the three or four coordinates of $\boldsymbol{\psi}$, a trapezoidal rule in $\operatorname{atanh}\phi$ with a Gauss–Hermite rule over the remaining coordinates at each node. It is very accurate for two reasons. The integrand at each node is the integrated likelihood from the grid filter, which is accurate to 10^{-11}, so the only error is that of the outer quadrature. Since the outer quadrature is over a few smooth coordinates, refining its rules changes the value by less than 10^{-4} (see Appendix B). The proposed approach uses the same grid filter: the optimization routine gives $\widehat{\boldsymbol{\psi}}$, $\boldsymbol{\Lambda}$ and the derivatives in (4). The residual $\log m_T - \log m_1 - \Delta_2$ reported below is the negative of the approximation term in (7).

Finally, we consider the following priors: $\mu \sim \mathcal{N}(0, 10)$, $(1+\phi)/2 \sim \mathcal{B}(20, 1.5)$ and $\sigma^2 \sim \mathcal{IG}(2.5, 0.1)$ in every model; $\lambda \sim \mathcal{N}(0, 1)$ in the stochastic volatility in mean model; and $\log(\nu - 2) \sim \mathcal{N}(2, 1)$ in the Student-t model. The coordinates are $\boldsymbol{\psi} = (\mu, \operatorname{atanh}\phi, \log\sigma^2)'$, extended by $\psi_4 = \lambda$ or $\psi_4 = \log(\nu - 2)$ in the two models with $d = 4$ parameters, and the Jacobian of the transform is in $p_{\boldsymbol{\psi}}$.

4.2 Accuracy of the Proposed Estimator

We first report the accuracy of the proposed estimator under the simulation design discussed above. Specifically, for each model, persistence level and sample size, Figure 1 plots the Laplace error $\log(m_T/m_1)$ and the residual $\log(m_T/m_1) - \Delta_2$ after the second-order term, as medians and ranges over the replications. The errors are in the logarithm, so they are approximately relative errors in m_T. For example, 10^{-2} corresponds to an

error of 1% in the marginal likelihood and in any Bayes factor computed from it.

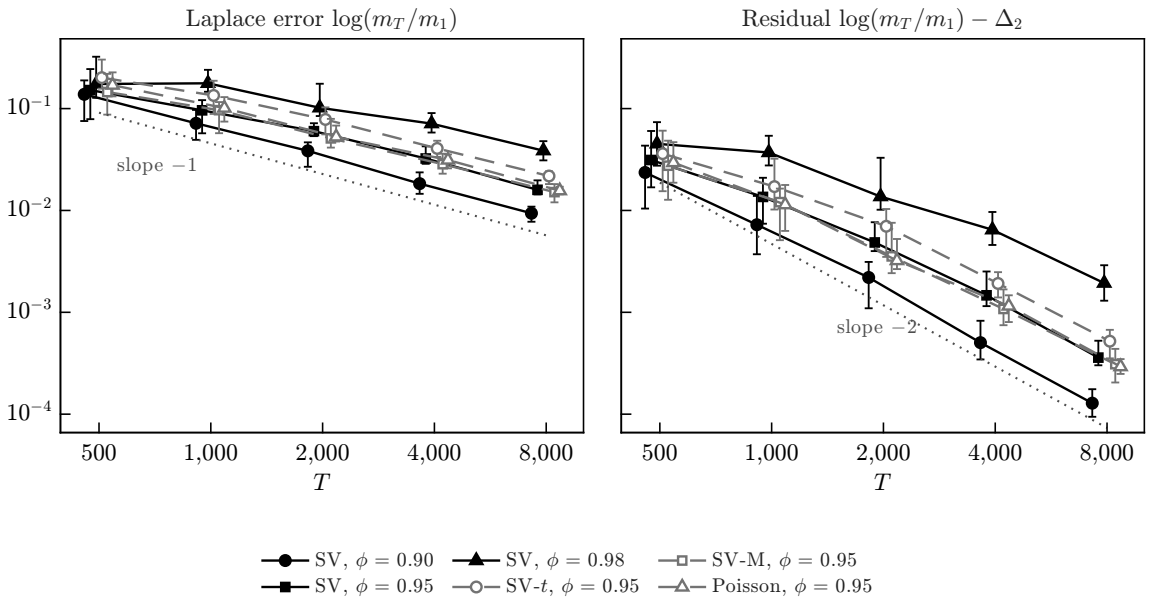


Figure 1: Laplace error $\log(m_T/m_1)$ (left) and residual after the second-order term $\log(m_T/m_1) - \Delta_2$ (right) against the sample size, on logarithmic scales. Markers are medians over the replications and bars their ranges, for the stochastic volatility model and the Student-t, stochastic volatility in mean and Poisson models. The dotted lines have slopes -1 and -2.

The left panel of Figure 1 shows that the Laplace approximation is reasonably accurate for every model and persistence level. Measured against the quadrature benchmark, it understates the log marginal likelihood by at most 0.33 at $T = 500$, 0.09 at $T = 4{,}000$ and 0.05 at $T = 8{,}000$. The error increases with the persistence, and the three related models behave like the stochastic volatility model at the same persistence.

The right panel shows that the second-order term removes between 64% and 99% of this error across the data sets. The residual falls roughly at the rate T^{-2} of Theorem 1, at which each doubling of T divides it by four. The second-order term also indicates the size of the residual: in 295 of the 300 data sets the residual lies between 1 and 4 times Δ_2^2, with median 1.5, so we use about $2\Delta_2^2$ as the indication of the remaining error.

4.3 Comparison with Simulation-Based Estimators

We now compare the proposed estimator with three commonly used marginal likelihood estimators on the stochastic volatility data sets in Section 4.1: importance sampling with the cross-entropy method of Chan and Eisenstat (2015), Chib's method (Chib, 1995), and the modified harmonic mean of Gelfand and Dey (1994) with the weighting density of Geweke (1999). All three require posterior draws from MCMC.

The three estimators share one posterior sampler, the auxiliary mixture sampler of Kim, Shephard, and Chib (1998). We obtain 20,000 posterior draws after a burn-in period of 5,000. The cross-entropy method fits a Gaussian proposal in $\boldsymbol{\psi}$ to these draws. It then evaluates the integrated likelihood at 10,000 draws from this proposal. Chib's method computes the posterior ordinate at the posterior mean of the draws, from the full run and two reduced runs. The Gelfand–Dey estimator evaluates the integrated likelihood at each of the 20,000 posterior draws, as Chan and Grant (2015) recommend for latent variable models. Because the auxiliary mixture sampler targets the mixture model, these two estimators reweight the draws to the exact model as in Kim, Shephard, and Chib (1998); the importance weights of the cross-entropy method correct for the mixture approximation directly. All four estimators use the grid filter for the integrated likelihood, so the comparison isolates the integration over the parameters.

To compare the estimators at equal cost, we rescale the numerical standard errors of the three simulation-based estimators to a computing budget of one minute on one core. We measure each standard error with the fixed numbers of draws given above, together with the computing time of each estimator on one core.[2] Since the number of draws is proportional to the computing time, a numerical standard error is inversely proportional to the square root of the computing time. Figure 2 plots these rescaled standard errors next to the error of the proposed estimator, by sample size and persistence level.

At this budget the error of the proposed estimator falls as T grows. The errors of Chib's method and the Gelfand–Dey estimator rise steeply with T, and that of the cross-entropy method rises by a factor of about two to five. The proposed estimator is therefore the most accurate of the four from $T = 1{,}000$ at $\phi = 0.90$ and from $T = 2{,}000$ at $\phi \geqslant 0.95$. At $T = 4{,}000$ its residual is 2 to 47 times smaller than the standard error of the cross-entropy method, the most accurate of the three simulation-based estimators, and at $T = 8{,}000$ it

[2]The computations of this section are in MATLAB. The timings are single-thread times on a laptop computer with an Intel Core Ultra 7 255U processor.

is 9 to 400 times smaller.

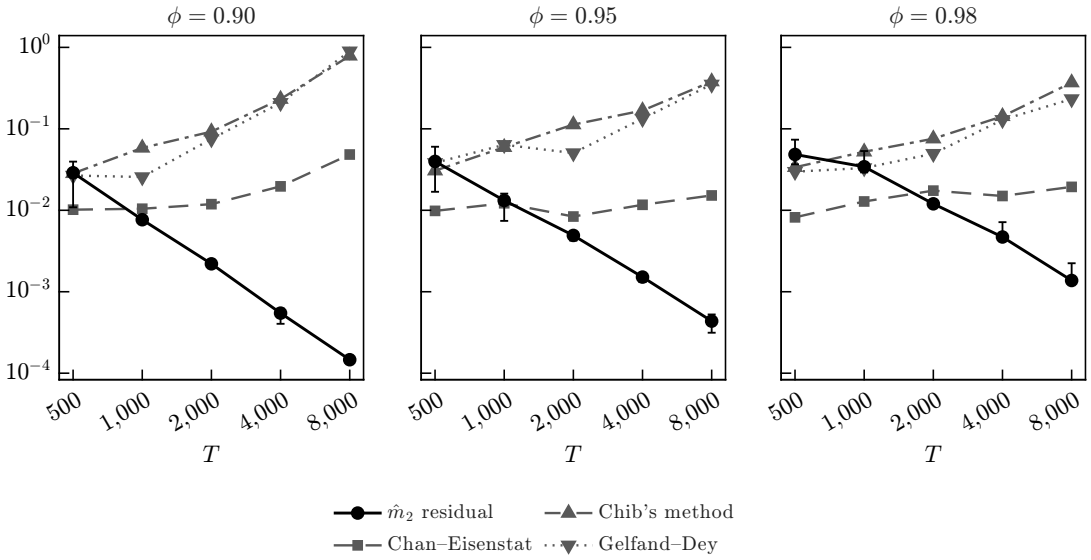


Figure 2: Error of the four marginal likelihood estimators (in log scale) at a computing budget of one minute on one core, against the sample size, for the stochastic volatility model at three persistence levels. For $\widehat{m}_2$ the error is the residual, as median and range over the replications; for the other estimators it is the numerical standard error scaled to the budget.

The simulation-based estimators lose accuracy as T grows for two reasons. First, each MCMC sweep and each likelihood evaluation takes longer at larger T. A fixed budget therefore allows fewer draws as T grows, and from $T = 4{,}000$ the three runs of Chib's method alone exceed it. The proposed estimator requires a number of likelihood evaluations that does not grow with T, and it takes 2.5 to 63 seconds, within the budget at $T \leqslant 4{,}000$ and about the budget at $T = 8{,}000$. Second, the reweighting of Chib's method and the Gelfand–Dey estimator to the exact model keeps a smaller share of the draws as T grows: at $T = 4{,}000$ the effective sample size is 0.1 to 3 percent of the draws, and at $T = 8{,}000$ it is below 0.5 percent. Their standard errors therefore grow with T even at a fixed number of draws, and at $T \geqslant 4{,}000$ their realized errors exceed these standard errors by up to a factor of six (see Appendix B). The cross-entropy method needs no reweighting. Its standard error at a fixed number of draws falls with T as the posterior becomes closer to Gaussian, which offsets much of the rising cost of each draw.

Overall, these results show that the proposed estimator is accurate for long time series at a small computing cost. Its error is roughly proportional to T^{-2}, and it takes about a minute or less on one core at every sample size. The simulation-based estimators need longer runs at larger T to reach a given accuracy. The advantage of the proposed estimator therefore grows with T.

4.4 Model Selection for Bivariate Stochastic Volatility

The Monte Carlo experiments above use scalar states, for which a quadrature benchmark exists. The models of the exchange-rate application have two- or three-dimensional states, for which no benchmark is available. We assess how often the proposed estimator selects the true model by marginal likelihood. We compare four of the bivariate stochastic volatility models of Yu and Meyer (2006), members of the multivariate stochastic volatility family reviewed by Asai, McAleer, and Yu (2006) and extended by Asai and McAleer (2009). These models are costly to estimate by MCMC. No auxiliary mixture samplers in the tradition of Kim, Shephard, and Chib (1998) have been developed for them. Yu and Meyer (2006) estimate these models using single-move samplers in WinBUGS, which Asai, McAleer, and Yu (2006) note can be numerically inefficient, and compare them by the deviance information criterion.

We write a pair of returns as $\mathbf{y}_t = (y_{1t}, y_{2t})'$ with $y_{it} = \mathrm{e}^{h_{it}/2}\varepsilon_{it}$, $i = 1, 2$. In the constant-correlation model (CC), the two log-volatilities follow independent AR(1) processes as in (1), and $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t})' \sim \mathcal{N}(\mathbf{0}, \boldsymbol{P})$, where $\boldsymbol{P}$ is a correlation matrix with off-diagonal element ρ. CC has a two-dimensional state and seven static parameters.

Each of the other three models changes one feature of CC. The Granger-causality model (GC) adds the lagged first log-volatility to the second transition, $h_{2,t+1} = \mu_2 + \phi_2(h_{2t} - \mu_2) + \phi_{21}(h_{1t} - \mu_1) + \sigma_2\eta_{2t}$. The heavy-tailed model (CC-t) replaces the Gaussian $\boldsymbol{\varepsilon}_t$ by a bivariate Student-t with scale matrix $\boldsymbol{P}$ and $\nu > 2$ degrees of freedom, so that $\mathrm{Var}(\boldsymbol{\varepsilon}_t) = \nu\boldsymbol{P}/(\nu - 2)$. The dynamic-correlation model (DC) replaces the constant ρ by $\rho_t = (\mathrm{e}^{q_t} - 1)/(\mathrm{e}^{q_t} + 1) = \tanh(q_t/2)$, as in Yu and Meyer (2006). Here q_t is a third AR(1) state with mean μ_q, persistence ϕ_q and innovation variance σ_q^2, and the model has nine static parameters.

We use the same priors as in Section 4.1 for $(\mu_i, \phi_i, \sigma_i^2)$ and $(\mu_q, \phi_q, \sigma_q^2)$. The other priors are $\rho \sim \mathcal{U}(-1, 1)$, $\phi_{21} \sim \mathcal{N}(0, 1)$ and $\log(\nu - 2) \sim \mathcal{N}(2, 1)$. The coordinates of CC are

$\boldsymbol{\psi} = (\mu_1, \operatorname{atanh}\phi_1, \log\sigma_1^2, \mu_2, \operatorname{atanh}\phi_2, \log\sigma_2^2, \operatorname{atanh}\rho)'$. GC adds $\psi_8 = \phi_{21}$, CC-t adds $\psi_8 = \log(\nu - 2)$, and DC replaces $\operatorname{atanh}\rho$ by $(\mu_q, \operatorname{atanh}\phi_q, \log\sigma_q^2)$.

We generate data from each of the four models, with the parameters at their posterior modes for the exchange-rate data of Section 5. For each model we generate 50 series of $T = 1{,}000$ observations, a quarter of the exchange-rate sample, which makes the models harder to distinguish. We fit all four models to every series. For CC, GC and CC-t the log marginal likelihood is $\log\widehat{m}_2$ from the grid filter. For DC the state is three-dimensional, and the estimate is $\log\widehat{m}_1$ from the particle filter with 30,000 particles. The selected model is the one with the largest estimate.

For each series, Figure 3 plots the log Bayes factors of the other three models against the true model. The true model is selected in 196 of the 200 series. The four exceptions are series generated from CC. DC is selected in three of them, by 0.25 to 1.10, and GC in one, by 0.66. These margins correspond to posterior odds of at most three to one. Each of the other three models includes CC as a special or limiting case: GC when $\phi_{21} = 0$, CC-t as $\nu \to \infty$, and DC as the innovation variance of q_t goes to zero. On data from CC, the more flexible models fit as well as CC. They have only one or two additional static parameters, so the penalty for their complexity is modest. The median log Bayes factors against CC are therefore only -4.1 for GC, -3.3 for CC-t and -2.1 for DC.

When the true model has a feature that the others lack, it is selected in every series. For data from GC, CC-t and DC, the median log Bayes factors against the true model range from -14.0 to -25.4, and the 95th percentiles are -5.5 or below.

These results show that the proposed estimator reliably identifies the true model among the four bivariate specifications. The four exceptions occur when the data are generated from the constant-correlation baseline, and a more flexible model is selected by a small margin. This reflects a property of the Bayes factor: when the true model is nested in a more flexible one, the evidence for the restricted model accumulates only slowly with the sample size. For data generated from any of the other three models, the proposed estimator selects the true model in every replication.

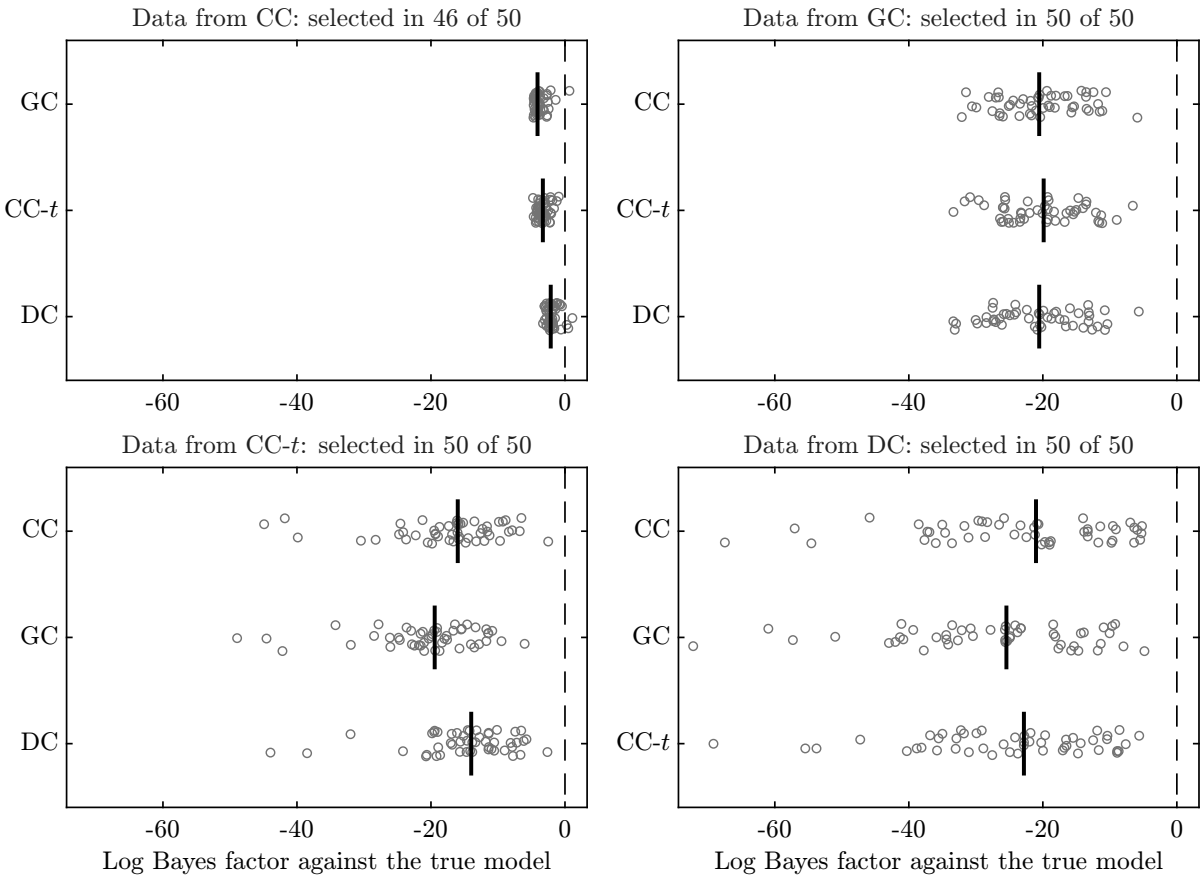


Figure 3: Log Bayes factors of the other three models against the true model on generated data with $T = 1{,}000$, one panel per data generating model. Each dot is one series, jittered vertically, and the bar is the median; values below zero favor the true model. The panel titles give the number of series in which the true model is selected.

5 Application: Bivariate Stochastic Volatility for Exchange Rates

This section applies the proposed approach to the four bivariate stochastic volatility models of Yu and Meyer (2006) for two exchange rates. The models have two- or three-dimensional states and seven to nine static parameters, and one likelihood evaluation takes several seconds. To our knowledge, the marginal likelihood has never been computed for these models. A marginal likelihood estimator that requires MCMC draws would need a posterior sampler for each model, and the Gelfand–Dey and cross-entropy estimators

would also need many more likelihood evaluations than the proposed approach. We first describe the data and model specifications, explain how their integrated likelihoods are evaluated, and then present the Bayes factors and computational costs.

5.1 Data and Models

We use daily log returns on the Australian and New Zealand dollars against the US dollar, constructed from the European Central Bank's daily euro reference rates. The sample spans 4,000 trading days from 16 February 2011 to 2 October 2026. Returns are expressed in percent and demeaned over the full ECB series, from January 1999 to October 2026. The two series have standard deviations of approximately 0.65 and 0.68 percent and a sample correlation of 0.83.

We compare the four bivariate stochastic volatility models introduced in Section 4.4. They share the observation equation $y_{it} = \mathrm{e}^{h_{it}/2}\varepsilon_{it}$ and first-order autoregressive log-volatility dynamics. The baseline constant-correlation model (CC) assumes Gaussian return shocks with correlation ρ. Each alternative extends the baseline in one direction: the Granger-causality model (GC) allows lagged AUD log-volatility to enter the NZD log-volatility equation; the heavy-tailed model (CC-t) replaces the Gaussian return shocks with a bivariate Student-t distribution with ν degrees of freedom; and the dynamic-correlation model (DC) introduces a third AR(1) state governing the correlation.

For CC, GC and CC-t the integrated likelihood is evaluated on a product grid in (h_{1t}, h_{2t}), and the derivatives in (4) come from the sparse design outlined in Section 2.3. For DC the state is three-dimensional, and the integrated likelihood comes from a bootstrap particle filter. Its log value is noisy, so the optimization fits a quadratic surface to the evaluations and the estimate is $\widehat{m}_1$.[3] Appendix B provides the details of the optimization settings.

[3]The transition kernel of the grid filter is the product of two one-dimensional banded kernels, or a block-banded kernel for GC. One evaluation takes between 4 and 14 seconds on one core at this T, and the evaluations of each optimization step run in parallel on 16 workers, in MATLAB on a desktop computer with an Intel Core i7-14700 processor. The sparse design has 770 points in seven dimensions and 1,168 in eight. The particle filter uses 30,000 particles, and its log value has a standard deviation of 0.73 at the mode.

5.2 Bayes Factors and Computing Cost

All three extensions receive strong support relative to the constant-correlation baseline, but heavy-tailed observations provide the largest improvement. As Table 1 shows, CC-t has a log Bayes factor of 110.3 against CC, followed by DC at 104.4 and GC at 97.7.

Table 1: Bivariate stochastic volatility models for the AUD/USD and NZD/USD returns, $T = 4{,}000$. The parameters are at the posterior mode, with $\rho = \tanh(\mu_q/2)$ and ϕ_q the persistence of q_t for DC. The column $\log \mathrm{BF}$ gives the log Bayes factor against CC, whose $\log \widehat{m}_2$ is $-5{,}474.86$; for GC and CC-t it is based on $\log \widehat{m}_2$ and for DC on $\log \widehat{m}_1$. Evaluations are the number of likelihood evaluations, by the particle filter for DC, and minutes are the wall-clock time on 16 workers.

Model	ϕ_1	ϕ_2	ρ	ϕ_{21}	ν	ϕ_q	Δ_2	$\log \mathrm{BF}$	Evaluations	Minutes
CC	0.982	0.984	0.836	–	–	–	0.15	0	1,976	12
GC	0.977	0.003	0.845	0.829	–	–	0.08	97.7	4,330	129
CC-t	0.985	0.987	0.837	–	7.9	–	0.18	110.3	3,266	32
DC	0.986	0.985	0.840	–	–	0.986	–	104.4	2,347	54

The parameter estimates help explain these gains. CC-t accommodates heavy tails with $\nu = 7.9$, while leaving the volatility parameters close to their baseline estimates. DC instead allows persistent movements in correlation, with a long-run level near 0.84 and persistence of 0.986. GC changes the dynamics of NZD volatility more substantially: the coefficient on its own lag falls to 0.003, while the coefficient on lagged AUD log-volatility is 0.83.

The higher-order correction changes the log Bayes factors among CC, GC and CC-t by less than 0.1, leaving their ranking unchanged. The comparison involving DC is less precise because it relies on the ordinary Laplace approximation and a particle-filter likelihood estimate.[4]

The computational cost remains manageable despite the models' latent-state dimensions. CC, GC and CC-t require between 2,000 and 4,300 likelihood evaluations per model, taking from 12 minutes to about two hours on 16 workers. DC requires 2,347 particle-filter

[4] Averaging 30 independent particle-filter runs at the mode gives a numerical standard error of 0.12 for the log integrated likelihood. This measures uncertainty in the final likelihood evaluation; it does not include uncertainty from estimating the mode and curvature using a fitted surface or error from omitting the higher-order correction.

evaluations and takes 54 minutes. MCMC-based alternatives would require posterior simulation for each specification, with additional reduced runs for Chib's method. For a Gelfand–Dey estimator based on the integrated likelihood, evaluating the likelihood at tens of thousands of posterior draws would itself be costly. At 4 to 14 seconds per evaluation, 20,000 evaluations would take about one to three days on a single core, compared with 2 to 17 hours of single-core time for the optimization underlying our approximation.

6 Application: Stochastic Volatility for Intraday Returns

Our second application examines whether extreme intraday returns are better explained by heavy-tailed observations, jumps, or short-lived movements in volatility. We compare seven stochastic volatility models using a sample of $T = 19{,}266$ five-minute returns on the SPDR S&P 500 ETF (SPY) over 2025. Four models have a scalar state: the standard stochastic volatility model and its extensions with heavy tails, with jumps in the returns, and with jumps in the returns and the volatility. Three have two volatility factors, with and without heavy tails or jumps. The models have seven to eleven static parameters, and one likelihood evaluation takes about a second or less for the scalar-state models and several seconds for the two-factor models. We first describe the data and the models, and then report the Bayes factors and the computing cost.

6.1 Data and Models

The data are five-minute log returns in percent on the SPDR S&P 500 ETF (SPY) in the regular session, 9:30 to 16:00, on the 247 full trading days of 2025. The price of each interval is its last trade and the price at 9:30 is the first trade of the day, so there are 78 returns per day and no overnight return. The returns are demeaned over the sample.[5]

The series has $T = 19{,}266$ returns, a standard deviation of 0.106 percent and a kurtosis of 153. The kurtosis is mostly due to the tariff news of April 2025. The two largest returns,

[5]The trades are from the TAQ millisecond consolidated trades file on Wharton Research Data Services (WRDS), 201.6 million in the year. We drop trades whose condition codes mark them as out of sequence, canceled, corrected or outside regular hours, and we exclude the three early-close days (July 3, November 28 and December 24).

3.82 and -2.93 percent, are ten minutes apart on the morning of April 7, and the standard deviation of the returns on that day is 0.73 percent, against 0.06 on the median day.[6] Without the four days from April 7 to 10, the kurtosis is 10.9. The autocorrelation of the squared returns is then 0.23 at lag 1, 0.26 at lag 5, 0.19 at lag 10 and 0.12 at lag 20, with a secondary peak of 0.11 at the daily lag of 78. The slow decay motivates a persistent volatility factor, and while the daily peak supports including a recurring time-of-day component. The extreme returns also motivate comparing heavy-tailed observations and jumps with a second, faster-moving volatility factor.

We compare seven specifications that accommodate these features in different ways. The baseline model is the stochastic volatility model in (1), denoted SV. The Student-t specification, SV-t, replaces its Gaussian return shock with $\varepsilon_t \sim \mathcal{T}_\nu$ with unit scale, as in Section 4.1. The remaining models introduce return jumps, volatility jumps, or a second volatility factor. These extensions build on specifications considered by Chernov, Gallant, Ghysels, and Tauchen (2003) for daily stock index returns.

The return-jump model, SVJ, adds an occasional shock to the observation equation, as in Chib, Nardari, and Shephard (2002):

$$y_t = \mathrm{e}^{h_t/2}\varepsilon_t + b_t\zeta_t,$$

where the jump indicator b_t equals one with probability ω and zero otherwise, the jump size is $\zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2)$, and both are independent over time and of the other shocks. Given h_t, the return is a mixture of two normal distributions with mean zero, variances e^{h_t} and $\mathrm{e}^{h_t} + \sigma_\zeta^2$, and weights $1-\omega$ and ω. The state remains the scalar h_t.

In the fourth model (SVJV) the same jump also enters the log-volatility, as in the model of Eraker, Johannes, and Polson (2003) with jumps in returns and volatility:

$$y_t = \mathrm{e}^{h_t/2}\varepsilon_t + b_t\zeta_t, \qquad h_t = \mu + \phi(h_{t-1} - \mu) + \sigma\eta_t + b_t\xi_t, \qquad \xi_t \sim \mathcal{N}(\mu_\xi, \sigma_\xi^2),$$

so that a jump in the return arrives together with a step in the log-volatility, which then decays at the rate ϕ. Given h_{t-1}, the pair (y_t, h_t) is a mixture of two components with weights $1-\omega$ and ω: the no-jump transition with the observation density $\mathcal{N}(0, \mathrm{e}^{h_t})$, and the transition shifted by ξ_t with the observation density $\mathcal{N}(0, \mathrm{e}^{h_t} + \sigma_\zeta^2)$. The state is again

[6] The morning of April 7 was marked by a sharp rally and reversal following false reports of a 90-day tariff pause and their subsequent denial by the White House. The actual tariff-pause announcement came on April 9, two days later.

the scalar h_t, with h_1 from the stationary distribution of the recursion without jumps. Through the shared jump the observation density depends on (h_{t-1}, h_t), and the grid filter sums over the two components at each date.

The last three models split the log-volatility into a persistent and a transitory factor:

$$y_t = \mathrm{e}^{(h_{1t}+h_{2t})/2}\varepsilon_t, \qquad h_{1t} = \mu + \phi_1(h_{1,t-1} - \mu) + \sigma_1\eta_{1t}, \qquad h_{2t} = \phi_2 h_{2,t-1} + \sigma_2\eta_{2t},$$

where η_{1t} and η_{2t} are independent standard normal and both factors start from their stationary distributions. The restriction $\phi_2 < \phi_1$ makes h_{1t} the persistent factor. The two-factor stochastic volatility model (TF) has a Gaussian ε_t, the two-factor Student-t model (TF-t) has $\varepsilon_t \sim \mathcal{T}_\nu$ with unit scale, and the two-factor jump model (TFJ) adds the jump $b_t\zeta_t$ of the third model to the return. Their state is two-dimensional, and the grid filter evaluates the integrated likelihood on a product grid.

All seven models account for the pronounced time-of-day pattern in intraday volatility (Andersen and Bollerslev, 1997). We add a deterministic component $\kappa(\tau_t)$ to the log-volatility in the observation equation, h_t or $h_{1t}+h_{2t}$. The term has the first two harmonics of the trading day,

$$\kappa(\tau_t) = \kappa_1\cos(2\pi\tau_t) + \kappa_2\sin(2\pi\tau_t) + \kappa_3\cos(4\pi\tau_t) + \kappa_4\sin(4\pi\tau_t),$$

where $\tau_t = j_t/78$ and $j_t \in \{1, \ldots, 78\}$ indexes the 78 five-minute intervals in each full trading day, with the index resetting at the start of each day. This component repeats each day and multiplies the variance of the non-jump return component by $\mathrm{e}^{\kappa(\tau_t)}$, leaving the latent factors to capture fluctuations around the recurring pattern. The coefficients $\kappa_1, \ldots, \kappa_4$ are estimated jointly with the dynamics, as in Stroud and Johannes (2014), so the Bayes factors do not depend on a pattern estimated beforehand. The term enters the observation density only, and the filters are unchanged.

We use the same priors on μ, ϕ, σ^2 and ν as in Section 4.1, with (ϕ_1, σ_1^2) in the role of (ϕ, σ^2) in the two-factor models. The jump probability has the prior $\omega \sim \mathcal{B}(2, 100)$, with mean 0.02, and the jump variance has the prior $\sigma_\zeta^2 \sim \mathcal{IG}(2.5, 0.1)$. The volatility jump has the priors $\mu_\xi \sim \mathcal{N}(0, 4)$ and $\sigma_\xi^2 \sim \mathcal{IG}(2.5, 0.1)$. In the two-factor models, $\phi_2 \,|\, \phi_1 \sim \mathcal{U}(-1, \phi_1)$ and $\sigma_2^2 \sim \mathcal{IG}(2.5, 0.1)$. The coefficients of the time-of-day term have the priors $\kappa_1, \ldots, \kappa_4 \sim \mathcal{N}(0, 1)$. We use the same coordinates as in Section 4.1, extended by $(\operatorname{logit}\omega, \log\sigma_\zeta^2)$ in the jump models, by $(\mu_\xi, \log\sigma_\xi^2)$ in the volatility-jump model and by

$(w, \log \sigma_2^2)$ in the two-factor models, where $w = \log(\operatorname{atanh} \phi_1 - \operatorname{atanh} \phi_2)$ imposes $\phi_2 < \phi_1$. The coefficients $\kappa_1, \ldots, \kappa_4$ are coordinates in every model, without transformation.

6.2 Bayes Factors and Computing Cost

For all seven models, the marginal likelihood estimator is $\log \widehat{m}_2$ from the grid filter. The two-factor models use a product grid in (h_{1t}, h_{2t}). The derivatives in (4) come from the sparse design of Section 2.3. Table 2 has the same layout as Table 1, with the posterior mode of the volatility and jump parameters and the log Bayes factor against SV.[7] The coefficients of the time-of-day term differ by less than 0.02 across the seven models and are omitted from the table; in SV they are $(\kappa_1, \kappa_2, \kappa_3, \kappa_4) = (0.30, 0.36, 0.16, 0.23)$. The fitted pattern rises from 2.9 times its minimum at the open to 3.5 times at about 10:00, falls to the minimum at 14:30 and rises again to 2.7 times at the close.

As shown in Table 2, heavy-tailed observations and return jumps improve the log marginal likelihood over SV by 25.1 and 12.3, respectively, but larger gains come from more flexible volatility dynamics: SVJV and TF improve on SV by 68.9 and 73.2. The parameter estimates help explain why these extensions are useful. Whereas SV captures volatility movements with a single persistence parameter of 0.987 and a half-life of 4.4 hours, TF separates a persistent component with $\phi_1 = 0.996$ and a half-life of about two trading days from a transitory component with $\phi_2 = 0.834$ and a half-life of nineteen minutes. The transitory component's innovation variance is ten times that of the persistent component, allowing large volatility movements to dissipate quickly.

SVJV also accommodates abrupt volatility increases, with an estimated jump probability implying approximately 131 expected jump events over the year and a mean log-volatility jump of 1.33. Their effects, however, decay with a half-life of about five hours. The log Bayes factor of 4.3 in favor of TF over SVJV suggests that the shorter-lived transitory component provides a better account of these movements.

Once the transitory factor is included, adding heavy tails or return jumps lowers the log marginal likelihood by 3.7 and 1.4, respectively. The corresponding parameter estimates also indicate less need for these features: the degrees of freedom rise from 15.9 in SV-t to 55.3 in TF-t, while the jump probability falls from 0.0035 in SVJ to 0.0010 in TFJ,

[7] The computations of this section are in MATLAB on a desktop computer with an Intel Core i9-12900KS processor. The evaluations of each optimization step run on eight workers.

implying approximately 67 and 20 expected jump events over the year. The higher-order correction has only a modest effect on the comparisons: its terms range from 0.02 to 0.31 and change the log Bayes factors by at most 0.3.

Table 2: Stochastic volatility models for the five-minute SPY returns, $T = 19{,}266$. The parameters are at the posterior mode; in the two-factor models, ϕ and σ^2 belong to the persistent factor and ϕ_2 and σ_2^2 to the transitory factor; ω is the jump probability per interval, σ_ζ^2 the variance of the return jump, and μ_ξ and σ_ξ^2 the mean and variance of the volatility jump. The column $\log \mathrm{BF}$ gives the log Bayes factor against SV, whose $\log \widehat{m}_2$ is 25,352.52. Evaluations are the number of likelihood evaluations, and minutes are the wall-clock time on eight workers.

Model	ϕ	σ^2	ϕ_2	σ_2^2	ν	ω	σ_ζ^2	μ_ξ	σ_ξ^2	Δ_2	$\log \mathrm{BF}$	Evaluations	Minutes
SV	0.987	0.040	–	–	–	–	–	–	–	0.02	0	2,794	3
SV-t	0.990	0.028	–	–	15.9	–	–	–	–	0.04	25.1	3,538	4
SVJ	0.988	0.034	–	–	–	0.0035	0.047	–	–	0.07	12.3	4,668	8
SVJV	0.989	0.019	–	–	–	0.0068	0.032	1.33	0.054	0.31	68.9	9,044	19
TF	0.996	0.010	0.834	0.098	–	–	–	–	–	0.07	73.2	5,342	57
TF-t	0.996	0.009	0.854	0.083	55.3	–	–	–	–	0.09	69.4	7,230	138
TFJ	0.996	0.010	0.847	0.087	–	0.0010	0.077	–	–	0.15	71.8	8,550	208

7 Concluding Remarks

This paper estimates the marginal likelihood of nonlinear state space models by a Laplace approximation over the static parameters, corrected by the second-order term of its expansion. The integrated likelihood is evaluated at the mode by an evaluator suited to the dimension of the state, and no posterior chain is run. For stochastic volatility models the corrected estimator is accurate to within 0.08 in the log marginal likelihood at $T = 500$, 0.01 at $T = 4{,}000$ and 0.003 at $T = 8{,}000$. The applications compare four bivariate models for two exchange rates and seven models for intraday returns at a cost of minutes to hours per model, with no sampler for any of them. Heavy tails are the best supported extension of the constant-correlation model for the exchange rates, and a transitory volatility factor is decisively supported for the intraday returns.

There are two interesting directions for future research. The first is the vector-state case with a particle-filter evaluator, of which the DC model of Section 5 is one instance.

There the optimization has to find the mode and curvature from noisy evaluations, and the question is how its cost scales with the dimension of the state and of the parameters. The second is sequential use. The mode and curvature at T are a warm start at $T+1$, so the predictive likelihood m_{T+1}/m_T and the posterior model weights can be updated each period with a few likelihood evaluations. This would give real-time Bayesian model averaging and density forecast comparison for nonlinear state space models without a sampler for any model.

Declaration of Generative AI

We acknowledge the use of Claude Code and Codex for editing, brainstorming and checking the code. A previous version of the paper was reviewed by Refine.ink, and its suggestions are incorporated in the current draft. All AI-assisted suggestions were carefully reviewed, and we take full responsibility for the final content and accuracy of this paper.

Appendix A Proofs

Proof of Proposition 1. Write $Z = Z_T(\boldsymbol{\psi})$ and $\pi(\boldsymbol{\alpha}) = \mathrm{e}^{\mathcal{C}(\boldsymbol{\psi},\boldsymbol{\alpha})}/Z$. For smooth $g_1, \ldots, g_k$ let $\mathcal{K}(\mathbf{t}) = \log \mathbb{E}_{\boldsymbol{\psi}} \exp(\sum_i t_i g_i)$ be their joint cumulant generating function, so that $\mathrm{cum}_{\boldsymbol{\psi}}(g_1, \ldots, g_k) = \partial_{t_1} \cdots \partial_{t_k} \mathcal{K}(\mathbf{0})$. Differentiating under the integral sign,

$$D_{\mathbf{w}} \mathcal{K}(\mathbf{t}) = \widetilde{\mathbb{E}}_{\mathbf{t}}\Big[\sum_i t_i D_{\mathbf{w}} g_i + c_1 \Big] - \mathbb{E}_{\boldsymbol{\psi}}[c_1],$$

where $\widetilde{\mathbb{E}}_{\mathbf{t}}$ is the expectation under the density proportional to $\exp(\sum_i t_i g_i)\pi(\boldsymbol{\alpha})$, and $\widetilde{\mathbb{E}}_{\mathbf{t}}[q] = \partial_s \log \mathbb{E}_{\boldsymbol{\psi}} \exp(\sum_i t_i g_i + sq)|_{s=0}$ for any q. Applying $\partial_{t_1} \cdots \partial_{t_k}$ at $\mathbf{t} = \mathbf{0}$ gives (9). The term $t_i D_{\mathbf{w}} g_i$ contributes $\mathrm{cum}_{\boldsymbol{\psi}}(g_1, \ldots, D_{\mathbf{w}} g_i, \ldots, g_k)$, and the term c_1 contributes $\mathrm{cum}_{\boldsymbol{\psi}}(g_1, \ldots, g_k, c_1)$. Differentiating Z under the integral sign gives $D_{\mathbf{w}} \log Z = \mathbb{E}_{\boldsymbol{\psi}}[c_1]$ directly. Since $D_{\mathbf{w}} c_j = c_{j+1}$, repeated application of (9) gives

$$\begin{aligned}
D_{\mathbf{w}} \mathbb{E}_{\boldsymbol{\psi}}[c_1] &= \mathbb{E}_{\boldsymbol{\psi}}[c_2] + \mathrm{Var}_{\boldsymbol{\psi}}(c_1), \\
D_{\mathbf{w}} \mathrm{Var}_{\boldsymbol{\psi}}(c_1) &= 2\mathrm{Cov}_{\boldsymbol{\psi}}(c_1, c_2) + \mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_1), \\
D_{\mathbf{w}} \mathrm{Cov}_{\boldsymbol{\psi}}(c_1, c_2) &= \mathrm{Var}_{\boldsymbol{\psi}}(c_2) + \mathrm{Cov}_{\boldsymbol{\psi}}(c_1, c_3) + \mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_2), \\
D_{\mathbf{w}} \mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_1) &= 3\mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_2) + \mathrm{cum}_{\boldsymbol{\psi}}(c_1, c_1, c_1, c_1),
\end{aligned}$$

and (8) follows by summing. $\square$

Proof of Theorem 1. Let φ denote the $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ density, let $\rho_T = T^{1/8}$, and write $\beta_k = K_k T^{1-k/2}$. Take T large enough that $\rho_T \leqslant \varrho\sqrt{T}$ and $\sum_{k=3}^{6} \beta_k \rho_T^k / k! \leqslant \log 2$. By (3), $m_T/m_1 = I_1 + I_2 + I_3$, where I_1 is the integral of $\mathrm{e}^R\varphi$ over the ball $\|\mathbf{u}\| \leqslant \rho_T$, I_2 is the integral over the shell $\rho_T < \|\mathbf{u}\| \leqslant \varrho\sqrt{T}$, and $I_3 = m_1^{-1} \int_{\mathcal{N}_T^c} \mathrm{e}^f$. Assumption 3 gives $I_3 \leqslant C_0 T^{-2}$.

For $\|\mathbf{u}\| \leqslant \varrho\sqrt{T}$, the segment from $\mathbf{0}$ to $\mathbf{u}$ lies in $\mathcal{N}_T$ because $\mathcal{N}_T$ is convex. Taylor's theorem along this segment gives $R(\mathbf{u}) = D^3 f(\boldsymbol{\xi})[\mathbf{Au}, \mathbf{Au}, \mathbf{Au}]/6$ for some $\boldsymbol{\xi} \in \mathcal{N}_T$, so $|R(\mathbf{u})| \leqslant \beta_3 \|\mathbf{u}\|^3/6 \leqslant (\varrho K_3/6)\|\mathbf{u}\|^2 \leqslant \frac{1}{4}\|\mathbf{u}\|^2$ by Assumption 2. Hence $\mathrm{e}^{R(\mathbf{u})}\varphi(\mathbf{u}) \leqslant (2\pi)^{-d/2}\mathrm{e}^{-\|\mathbf{u}\|^2/4}$ on $\mathcal{N}_T$, and

$$I_2 \leqslant (2\pi)^{-d/2} \int_{\|\mathbf{u}\| > \rho_T} \mathrm{e}^{-\|\mathbf{u}\|^2/4} \, \mathrm{d}\mathbf{u} = \frac{2^{d/2}\,\Gamma(d/2, \rho_T^2/4)}{\Gamma(d/2)} \leqslant M_d \, \rho_T^{d-2} \, \mathrm{e}^{-\rho_T^2/4}$$

for a constant M_d, where $\Gamma(\cdot,\cdot)$ is the upper incomplete gamma function. This bound is $o(T^{-2})$.

On the ball, Taylor's theorem to sixth order gives $R = C_3 + C_4 + C_5 + R_6$. Here $|C_k(\mathbf{u})| \leqslant \beta_k \|\mathbf{u}\|^k/k!$ for every $\mathbf{u}$ by Assumption 2 at $\boldsymbol{\psi} = \widehat{\boldsymbol{\psi}}$, and $|R_6(\mathbf{u})| \leqslant \beta_6\|\mathbf{u}\|^6/720$ on the ball by the same assumption on $\mathcal{N}_T$. On the ball $|R(\mathbf{u})| \leqslant \sum_{k=3}^{6} \beta_k \rho_T^k/k! = O(T^{-1/8})$, and $\mathrm{e}^R = \sum_{j=0}^{4} R^j/j! + E_5$ with $|E_5| \leqslant |R|^5 \mathrm{e}^{|R|}/120 \leqslant |R|^5/60$. Each monomial in the expansion of $\sum_{j\leqslant 4} R^j/j!$, and the bound on E_5, is dominated by a constant times a product of factors $\beta_{k_i}\|\mathbf{u}\|^{k_i}$. The integral of such a product against φ over the ball is at most the corresponding Gaussian moment over $\mathbb{R}^d$, which is finite, times $\prod_i \beta_{k_i} = O(T^{\sum_i (1-k_i/2)})$. The terms of order $T^{-1/2}$ and $T^{-3/2}$ are C_3, C_5, C_3C_4 and $C_3^3/6$. They are odd functions of $\mathbf{u}$ and integrate to zero over the ball. The terms of order T^{-1} are C_4 and $\frac{1}{2}C_3^2$. Every other term, including E_5, is of order T^{-2} or smaller. Therefore

$$I_1 = \int_{\|\mathbf{u}\| \leqslant \rho_T} \big\{1 + C_4(\mathbf{u}) + \tfrac{1}{2}C_3(\mathbf{u})^2\big\}\varphi(\mathbf{u}) \, \mathrm{d}\mathbf{u} + O(T^{-2}),$$

and extending the integral to $\mathbb{R}^d$ changes it by at most a constant times $(1 + \beta_4\rho_T^4 + \beta_3^2\rho_T^6)\rho_T^{d-2}\mathrm{e}^{-\rho_T^2/2}$, which is $o(T^{-2})$. The Gaussian moments give $\mathbb{E}[C_4(\mathbf{G})] = \frac{1}{8}\sum_{i,j} b_{iijj}$, because $\mathbb{E}[G_iG_jG_kG_l]$ is the sum over the three pairings of the indices. They also give $\frac{1}{2}\mathbb{E}[C_3(\mathbf{G})^2] = \frac{1}{12}\sum_{i,j,k} a_{ijk}^2 + \frac{1}{8}\sum_i (\sum_j a_{ijj})^2$, because $\mathbb{E}[G_iG_jG_kG_lG_mG_n]$ is the sum over the fifteen pairings, of which six pair each index of one cubic factor with an index of the

other and nine pair one index within each factor. Hence $m_T/m_1 = 1 + \Delta_2 + O(T^{-2})$. By polarization, i.e., the identity that expresses a symmetric multilinear form through its diagonal values, Assumption 2 at $\boldsymbol{\psi} = \widehat{\boldsymbol{\psi}}$ bounds every entry of the standardized tensors, so $\Delta_2 = O(T^{-1})$, and $\log(1 + \Delta_2 + r) = \Delta_2 + O(T^{-2})$ for $r = O(T^{-2})$. All constants depend only on d, ϱ, $K_3, \dots, K_6$ and C_0. □

Proof of Proposition 2. Unbiasedness and the variance bound are Corollary 1 of Chan (2026): $\mathbb{E}[U_m] = p_{\mathrm{a}}$ and $0 < U_m \leqslant 1$ give $U_m^2 \leqslant U_m$ and $\mathrm{Var}(U_m) \leqslant p_{\mathrm{a}}(1-p_{\mathrm{a}})$. Theorem 1 of Hoeffding (1963) states, for independent variables with values in $[0,1]$ and mean p_{a}, that $\mathbb{P}\{\bar{U}_M \geqslant p_{\mathrm{a}} + t\} \leqslant \exp\{-M\,\mathrm{kl}(p_{\mathrm{a}} + t \| p_{\mathrm{a}})\}$ for $0 < t < 1 - p_{\mathrm{a}}$. Applying it to $1 - U_m$ gives the same bound for the lower tail. Let $\varepsilon = \log(2/\gamma)/M$. The function $p \mapsto \mathrm{kl}(p \| p_{\mathrm{a}})$ is continuous, zero at $p = p_{\mathrm{a}}$ and increasing on $[p_{\mathrm{a}}, 1]$. The event $\{\bar{U}_M > p_{\mathrm{a}},\ \mathrm{kl}(\bar{U}_M \| p_{\mathrm{a}}) > \varepsilon\}$ is therefore contained in $\{\bar{U}_M \geqslant p_{\mathrm{a}} + t\}$ with t the solution of $\mathrm{kl}(p_{\mathrm{a}} + t \| p_{\mathrm{a}}) = \varepsilon$, or is empty when no solution exists in $(0, 1 - p_{\mathrm{a}})$. Its probability is at most $\mathrm{e}^{-M\varepsilon} = \gamma/2$. The lower tail is the same. □

Appendix B Implementation Details

This appendix gives the implementation details behind the results in Sections 4 to 6. We first describe the two evaluators of the integrated likelihood for the scalar-state models and the quadrature benchmark, then the mode search and the sparse design for the derivatives in Δ_2. The last two subsections cover the MCMC-based marginal likelihood estimators of Section 4.3 and the parameter transformations with their Jacobians.

B.1 Integrated Likelihood Evaluators and Benchmark

We first describe the grid filter for the models in Section 4, whose state is scalar with transition $h_t = \mu + \phi(h_{t-1} - \mu) + \sigma\eta_t$. At a given $\boldsymbol{\theta}$, Newton's method with the banded precision matrix gives the smoothing mode $\widehat{\mathbf{h}}$ and the marginal standard deviations $\widehat{s}_t$ of the Gaussian approximation to $p(\mathbf{h} \,|\, \mathbf{y}, \boldsymbol{\theta})$. The grid has spacing $\delta = \min(\sigma/2, 0.05)$ and runs from $\min_t \widehat{h}_t - 12 \max_t \widehat{s}_t - 6\sigma$ to $\max_t \widehat{h}_t + 12 \max_t \widehat{s}_t + 6\sigma$. With a spacing of at most half the transition standard deviation, the aliasing error of the trapezoidal rule for the Gaussian kernel is at most of order $\mathrm{e}^{-8\pi^2}$. The transition matrix $\mathbf{K}$ of Section 2.2 is stored as a sparse matrix with the entries beyond 8.7σ from the conditional mean set to zero.

The product grids of the bivariate models in Section 5 have spacing $\min(\sigma_i/2, 0.1)$ in each state, and those of the two-factor models in Section 6 have spacing $0.6\sigma_i$, at which the aliasing error is still below 10^{-23}. One evaluation of $\log Z_T$ takes between 0.01 seconds at $T = 500$ and 0.19 seconds at $T = 8{,}000$ on one core. For a linear Gaussian model, whose integrated likelihood is available in closed form, the filter agrees with the exact value to 3×10^{-12}. On the 300 data sets in Section 4, it also agrees with the bounded-weight evaluator with $M = 1{,}000$ proposals: the standardized difference between the two values of $\log Z_T(\widehat{\boldsymbol{\psi}})$ has mean 0.11 and standard deviation 1.08, and the largest difference is 0.001 in $\log Z_T$, at $T = 8{,}000$.

The bounded-weight evaluator of Section 2.2 uses the tangent-twisted proposal of Chan (2026), with $G = 41\sqrt{T/500}$ nodes per date, rounded to the nearest integer, at the smoothing mode plus equally spaced multiples of the smoothing standard deviation on $[-4, 6]$. One proposal is a forward pass through the twisted transitions. Its log weight is the sum over dates of the tangent-envelope deficits, which is nonpositive by construction, so $U_m = w(\boldsymbol{\alpha}^{(m)})/C(\widehat{\boldsymbol{\psi}}) \in (0, 1]$.

The benchmark value of $\log m_T$ in Section 4 is a nested quadrature in $\boldsymbol{\psi}$. The outer rule is a trapezoidal rule in $z = \operatorname{atanh} \phi$ with step 0.5 posterior standard deviations, extended until the integrand falls below 10^{-9} of its maximum. The integrand is analytic in z with Gaussian-like tails, so the trapezoidal rule converges exponentially and the error of the outer rule is far smaller than the truncation at 10^{-9}. The inner rule is a Gauss–Hermite rule over the remaining coordinates with 8 points per dimension (6 when the inner integral is three-dimensional), centered at the conditional mode and scaled by the conditional curvature. Refining both rules changes $\log m_T$ by at most 4.2×10^{-5} on the 240 data sets with $T \leqslant 4{,}000$, and by at most 3×10^{-9} on the nine stochastic volatility data sets of the comparison at $T = 8{,}000$. For the Student-t model at $T \leqslant 2{,}000$, where the conditional posterior of ψ_4 has a long right tail, the inner rule in ψ_4 is a trapezoidal rule.

B.2 Mode Search and Sparse Design

The mode search uses Newton steps in the standardized coordinates $\mathbf{u} = \mathbf{A}^{-1}(\boldsymbol{\psi} - \widehat{\boldsymbol{\psi}})$, with the gradient and Hessian from central differences.[8] Each step is evaluated at nine

[8] Fourth-order central differences of step 0.1, with Richardson extrapolation for the mixed second derivatives. The nine scales are $0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8$.

scales in one parallel batch and the best is taken, because for some models the full Newton step is too short along a curved ridge of the posterior, as for GC. Where the Hessian is not negative definite, the step follows the gradient. In Section 6 it takes the modified Newton direction described below instead. The iteration stops when the standardized step is below 10^{-7} in the experiments and 10^{-6} in the applications.

The derivatives in (4) come from central differences of $R(\mathbf{u})$ at $\mathbf{u} = \mathbf{0}$ on a sparse design with step $\ell = 0.25$: the points $\pm\ell\mathbf{e}_i$ and $\pm2\ell\mathbf{e}_i$ on each axis give a_{iii} and b_{iiii}, the four points $\pm\ell\mathbf{e}_i \pm \ell\mathbf{e}_j$ in each coordinate plane give a_{iij}, a_{ijj} and b_{iijj}, and the eight points $\pm\ell\mathbf{e}_i \pm \ell\mathbf{e}_j \pm \ell\mathbf{e}_k$ in each coordinate cube give a_{ijk}. The design is sparse because (4) needs only the third derivatives and the fourth derivatives with paired indices, so its points lie on the coordinate axes, planes and cubes, $6d + 8\binom{d}{2} + 16\binom{d}{3}$ in all, against 5^d for a full central-difference grid. These differences have an error of order ℓ^2, which under Assumption 2 changes Δ_2 by $O(T^{-2})$, the order of the remainder in Theorem 1; repeating them at step 2ℓ and taking $(4D_\ell - D_{2\ell})/3$ for each derivative D removes this leading term (Burden and Faires, 2011, Section 4.2). In the experiments the optimization requires between 258 and 760 evaluations per data set, of which the design takes 58 for $d = 3$ and 136 for $d = 4$.

In the two-factor stochastic volatility models in Section 6 the Hessian is indefinite over most of the path from the start, along a ridge in $(\operatorname{atanh}\phi_1, w)$, where gradient steps advance slowly. The search therefore takes a modified Newton step there: the eigenvalues of the standardized Hessian are replaced by the negatives of their absolute values, clipped to $[-4, -0.25]$, and the step is capped at two standardized units. The three two-factor searches converge in 10, 11 and 10 iterations.

For the DC model, the evaluations of the integrated likelihood are noisy, so the mode search works with fitted surfaces instead of Newton steps, in the manner of the response surface method of Box and Wilson (1951). At each round we draw design points at random around the current center, evaluate the log posterior kernel at them with the particle filter under common random numbers, as in the smooth particle filter of Malik and Pitt (2011). We fit a quadratic surface to these values by a robust regression, so that a few points far below the surface do not drive the fit.[9] We then move the center by

[9] At each round the design is the center and 330 points drawn from a normal distribution in the standardized coordinates, with standard deviation 1.5 in the first two rounds and 1 afterwards, and the surface is fitted with Huber and then bisquare weights. The fitted curvatures are clipped to $[0.25, 4]$, which limits the change of the scales to a factor of two per round, and the Newton step is capped at 1.5

the capped Newton step of the fitted surface and rescale the coordinates by its clipped curvature. The search starts at the CC mode and stops when two consecutive fits are concave with short Newton steps, so that the curvature in $\widehat{m}_1$ comes from a fit centered near the mode.

B.3 MCMC-Based Marginal Likelihood Estimators

Chib's method, the Gelfand–Dey estimator and the cross-entropy method require posterior draws. For the stochastic volatility model in Section 4.3, we obtain them with the auxiliary mixture sampler of Kim, Shephard, and Chib (1998) in conjunction with the precision-based draw of the volatility path (Chan and Jeliazkov, 2009). The sampler uses the seven-component mixture, Gibbs steps for μ and σ^2, and the independence Metropolis–Hastings step of Kim, Shephard, and Chib (1998) for ϕ. The full run starts from $\phi = 0.95$, $\sigma^2 = 0.1$ and μ equal to the logarithm of the sample variance.

Chib's method evaluates the posterior ordinate at the posterior mean of the full run: the ordinate of μ comes from the full run, that of σ^2 from a reduced run with μ fixed, and that of ϕ from a second reduced run, normalized by quadrature. The Gelfand–Dey estimator uses the truncated normal of Geweke (1999) at the 95% ellipsoid in $\boldsymbol{\psi}$, and the cross-entropy method a Gaussian in $\boldsymbol{\psi}$ with the mean and covariance of the draws of the full run. Because the auxiliary mixture sampler targets the mixture approximation, Chib's method and the Gelfand–Dey estimator reweight each draw to the exact model. The weight is $\prod_{t=1}^{T} p(\varepsilon_t^*)/q(\varepsilon_t^*)$, where $\varepsilon_t^* = \log y_t^2 - h_t$, p is its exact $\log \chi_1^2$ density and q the seven-component mixture. The cross-entropy method uses the integrated likelihood of the exact model and needs no reweighting.

Numerical standard errors are computed by batch means with 20 batches. The computing times in Section 4.3 are single-thread times on an idle core, extrapolated from the time of one sweep of each run and of one likelihood evaluation at each T and ϕ. At $T = 4{,}000$, where the reweighting keeps 0.1 to 3 percent of the draws, the largest errors are 0.76 for Chib's method and 0.13 for the Gelfand–Dey estimator, 6.2 and 4.5 times their standard errors. At $T = 8{,}000$ the reweighting keeps at most 0.5 percent of the draws, and the largest errors are 1.45 and 0.37, with standard errors of up to 1.7 and 0.11. The errors

standardized units, or at 1 when the fit is not concave. The initial scales are the CC posterior standard deviations, the search stops when the Newton step is at most 0.5 standardized units, and $\widehat{Z}_T(\widehat{\boldsymbol{\psi}})$ is the average of 30 independent filters at the mode.

of the cross-entropy method are at most 0.020 at every T.

B.4 Parameter Transformations and Jacobians

This subsection records the transformations that map the parameters of each model to the unconstrained coordinates $\boldsymbol{\psi}$, and the terms that they add to $\log p_{\boldsymbol{\psi}}$. For a scalar parameter θ with prior density $p(\theta)$ and coordinate $\psi = g(\theta)$, the prior density of ψ is $p(g^{-1}(\psi))\,|\mathrm{d}g^{-1}(\psi)/\mathrm{d}\psi|$. The coordinates of each model are listed in Sections 4.1, 4.4 and 6.1, and $\log p_{\boldsymbol{\psi}}$ is the sum over the coordinates of the terms below. Six cases cover all the models.

A parameter with a normal prior enters as itself and adds no Jacobian: the levels μ, μ_1, μ_2 and μ_q, the coefficients λ and ϕ_{21}, the time-of-day coefficients $\kappa_1, \ldots, \kappa_4$, the mean μ_ξ of the volatility jump, and the degrees of freedom through $\psi = \log(\nu - 2)$, on which the prior $\mathcal{N}(2, 1)$ is placed directly. Under this prior $\nu - 2$ is log-normal, with density

$$p(\nu) = \frac{1}{(\nu - 2)\sqrt{2\pi}} \exp\left\{-\frac{(\log(\nu - 2) - 2)^2}{2}\right\}, \qquad \nu > 2,$$

whose median is $2 + \mathrm{e}^2 \approx 9.4$ and whose mean is $2 + \mathrm{e}^{5/2} \approx 14.2$.

A persistence parameter with prior $(1 + \phi)/2 \sim \mathcal{B}(a, b)$ enters as $z = \operatorname{atanh}\phi$. Since $1 + \tanh z = \mathrm{e}^z/\cosh z$, $1 - \tanh z = \mathrm{e}^{-z}/\cosh z$ and $\mathrm{d}\phi/\mathrm{d}z = \cosh^{-2} z$,

$$\log p(z) = (a - b)z - (a + b)\log\cosh z - (a + b - 1)\log 2 - \log B(a, b),$$

where $B(a, b)$ is the beta function. This case covers ϕ, ϕ_1, ϕ_2 and ϕ_q in the models with independent persistence parameters.

A variance with prior $\sigma^2 \sim \mathcal{IG}(\nu_0, S_0)$, whose density is $S_0^{\nu_0}\Gamma(\nu_0)^{-1}(\sigma^2)^{-\nu_0 - 1}\mathrm{e}^{-S_0/\sigma^2}$, enters as $s = \log\sigma^2$. Since $\mathrm{d}\sigma^2/\mathrm{d}s = \mathrm{e}^s$,

$$\log p(s) = \nu_0 \log S_0 - \log\Gamma(\nu_0) - \nu_0 s - S_0\mathrm{e}^{-s}.$$

This case covers σ^2, σ_1^2, σ_2^2, σ_q^2, σ_ζ^2 and σ_ξ^2.

The correlation with prior $\rho \sim \mathcal{U}(-1, 1)$ enters as $r = \operatorname{atanh}\rho$, with $\mathrm{d}\rho/\mathrm{d}r = \cosh^{-2} r$, so $\log p(r) = -\log 2 - 2\log\cosh r$.

The jump probability with prior $\omega \sim \mathcal{B}(a_\omega, b_\omega)$ enters as $k = \operatorname{logit} \omega$. Since $\mathrm{d}\omega/\mathrm{d}k = \omega(1-\omega)$, $\log \omega = k - \log(1+\mathrm{e}^k)$ and $\log(1-\omega) = -\log(1+\mathrm{e}^k)$,

$$\log p(k) = a_\omega k - (a_\omega + b_\omega)\log(1+\mathrm{e}^k) - \log B(a_\omega, b_\omega).$$

The second persistence parameter of the two-factor models, with prior $\phi_2 \,|\, \phi_1 \sim \mathcal{U}(-1, \phi_1)$ and density $1/(1+\phi_1)$, enters as $w = \log(z_1 - z_2)$ with $z_i = \operatorname{atanh} \phi_i$, so that $\phi_2 = \tanh(z_1 - \mathrm{e}^w)$ and $\partial\phi_2/\partial w = -\mathrm{e}^w \cosh^{-2}(z_1 - \mathrm{e}^w)$. The conditional prior density of w given z_1 is

$$\log p(w \,|\, z_1) = w - 2\log\cosh(z_1 - \mathrm{e}^w) - \log(1 + \tanh z_1),$$

and z_1 itself carries the term of the persistence case above.

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